

Chroma ATE Inc. 2360.TW 2360 TT

EQUITY: TECHNOLOGY

Full power on the bright AI avenue

Riding on upgrades in AI/HPC chips and server power, with CPO test joining the lineup; resume at Buy

Resume coverage at Buy with TP of TWD2,845, implying ~34% upside

We resume coverage of Chroma with a Buy rating and a target price of TWD2,845, based on 38x average 2027-28F EPS of TWD74.9. Our target P/E multiple is at the higher end of Chroma's historical trading band of 14-58x, which we view as reasonable given a 39% net earnings CAGR through 2026-28F. The stock trades at 32x 2027F basic EPS of TWD65.6, compared to semi backend equipment/test interface vendors (Bloomberg consensus 2027E average P/E: 37x) and data center power peers (average: 27x). Chroma is a testing equipment maker primarily serving semi/photonics (38% of 2026F consolidated revenue) and power electronics (38%) customers. In semi/photonics, we expect Chroma to ride on the secular testing tailwinds from increasing design complexity and larger footprints particularly in AI/HPC chips, as well as the emergence of co-packaged optics (CPO) test insertions as a longer-term driver. In test instrument & automatic testing system (ATS), we believe Chroma is well-positioned to monetize the transition of the AI server power architecture toward high-voltage direct current (HVDC; $\pm 400V$ or 800V) power racks. We expect Chroma to record a revenue CAGR of 32% and model basic EPS of TWD43.6/TWD65.6/TWD84.2 in 2026F/27F/28F. Major downside risks to our view are CoWoS and backend testing capacity expansion slowdown and delayed HVDC deployments.

Handler upgrades and broader SLT adoption underway

We believe Chroma's system-level test (SLT) handler will enjoy a regular upgrade cycle given AI/HPC platform refresh cadences, and will particularly capitalize on the ASP uptrend from nVidia's (NVDA US, Not rated) adoption of GPU-on-GPU SoIC in 2028E for more stringent thermal control. In our view, TSMC (2330 TT, Buy) and OSATs' more aggressive 2.5D/CoWoS capacity investments and more spending on testing by AI OSATs could be positive indicators for Chroma's SLT business, and we see potentially more SLT adoption by AI/HPC chips in the longer run. We expect a 68% revenue CAGR for Semi/Photonics over 2025-28F.

Intact HVDC adoption, and Chroma benefits from more complicated testing

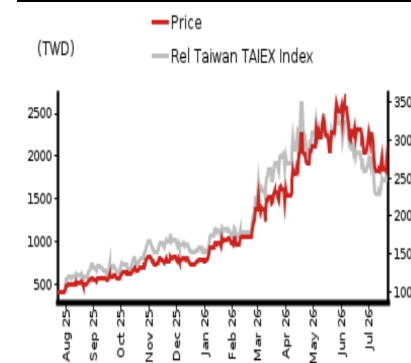
Despite recent market concerns about slower-than-expected HVDC adoption, we believe the adoption trajectory remains intact. A top US CSP is en-route to adopting $\pm 400VDC$ from 2H26F, likely beginning with VR200 racks before extending the architecture to its own ASIC racks. More sophisticated testing efforts, in our view, remain a key positive for Chroma, and we model an ATS revenue CAGR of 43% over 2025-28F.

Year-end 31 Dec	FY25	FY26F		FY27F		FY28F	
Currency (TWD)	Actual	Old	New	Old	New	Old	New
Revenue (mn)	28,311	0	53,514	0	74,977	0	93,623
Reported net profit (mn)	11,692	0	18,471	0	27,778	0	35,687
Normalised net profit (mn)	11,692	0	18,471	0	27,778	0	35,687
FD normalised EPS	27.59		43.42		65.30		83.90
FD norm. EPS growth (%)	121.7		57.4		50.4		28.5
FD normalised P/E (x)	77.0	–	48.9	–	32.5	–	25.3
EV/EBITDA (x)	83.7	–	37.3	–	25.0	–	19.2
Price/book (x)	28.2	–	21.3	–	15.7	–	12.3
Dividend yield (%)	0.9	–	1.4	–	2.2	–	2.8
ROE (%)	41.2		49.8		55.8		54.6
Net debt/equity (%)	net cash		net cash		net cash		net cash

Source: Company data, Nomura estimates

Rating	Buy
From Suspended	
Target price	TWD 2,845.00
From N/A	
Closing price	TWD 2,125.00
22 July 2026	
Implied upside	+33.9%
Market Cap (USD mn)	27,951.6
ADT (USD mn)	217.9

Relative performance chart



Source: LSEG, Nomura

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Key data on Chroma ATE Inc.

Performance

(%)	1M	3M	12M		
Absolute (TWD)	-8.0	4.7	400.0	M cap (USDmn)	27,951.6
Absolute (USD)	-9.9	1.9	355.9	Free float (%)	81.9
Rel to Taiwan	-0.7	-12.1	307.6	3-mth ADT (USDmn)	217.9
TAIEX Index					

Income statement (TWDmn)

Year-end 31 Dec	FY24	FY25	FY26F	FY27F	FY28F
Revenue	21,604	28,311	53,514	74,977	93,623
Cost of goods sold	-8,858	-10,886	-20,098	-27,806	-34,006
Gross profit	12,746	17,425	33,416	47,171	59,617
SG&A	-7,264	-8,227	-10,957	-13,491	-16,232
Employee share expense	0	0	0	0	0
Operating profit	5,482	9,198	22,458	33,680	43,385
EBITDA	6,257	10,038	23,429	35,077	45,390
Depreciation	-775	-841	-971	-1,398	-2,005
Amortisation	0	0	0	0	0
EBIT	5,482	9,198	22,458	33,680	43,385
Net interest expense	48	56	48	65	96
Associates & JCEs	669	734	759	1,022	1,143
Other income	510	3,930	155	0	0
Earnings before tax	6,709	13,918	23,421	34,767	44,623
Income tax	-1,308	-1,992	-4,686	-6,749	-8,696
Net profit after tax	5,400	11,926	18,735	28,018	35,927
Minority interests	-136	-234	-264	-240	-240
Other items	0	0	0	0	0
Preferred dividends	0	0	0	0	0
Normalised NPAT	5,264	11,692	18,471	27,778	35,687
Extraordinary items	0	0	0	0	0
Reported NPAT	5,264	11,692	18,471	27,778	35,687
Dividends	-3,828	-8,292	-12,930	-19,445	-24,981
Transfer to reserves	1,436	3,400	5,541	8,333	10,706

Valuations and ratios

Reported P/E (x)	170.1	76.7	48.7	32.4	25.2
Normalised P/E (x)	170.1	76.7	48.7	32.4	25.2
FD normalised P/E (x)	170.8	77.0	48.9	32.5	25.3
Dividend yield (%)	0.4	0.9	1.4	2.2	2.8
Price/cashflow (x)	187.9	171.3	69.7	47.1	26.7
Price/book (x)	36.2	28.2	21.3	15.7	12.3
EV/EBITDA (x)	130.5	83.7	37.3	25.0	19.2
EV/EBIT (x)	146.9	90.8	38.9	26.0	20.1
Gross margin (%)	59.0	61.5	62.4	62.9	63.7
EBITDA margin (%)	29.0	35.5	43.8	46.8	48.5
EBIT margin (%)	25.4	32.5	42.0	44.9	46.3
Net margin (%)	24.4	41.3	34.5	37.0	38.1
Effective tax rate (%)	19.5	14.3	20.0	19.4	19.5
Dividend payout (%)	72.7	70.9	70.0	70.0	70.0
ROE (%)	22.5	41.2	49.8	55.8	54.6
ROA (pretax %)	19.7	26.8	47.7	52.7	55.4

Growth (%)

Revenue	15.7	31.0	89.0	40.1	24.9
EBITDA	15.5	60.4	133.4	49.7	29.4
Normalised EPS	32.3	121.7	57.4	50.4	28.5
Normalised FDEPS	32.3	121.7	57.4	50.4	28.5

Source: Company data, Nomura estimates

Cashflow statement (TWDmn)

Year-end 31 Dec	FY24	FY25	FY26F	FY27F	FY28F
EBITDA	6,257	10,038	23,429	35,077	45,390
Change in working capital	-5,568	-3,098	-5,964	-9,183	-2,906
Other operating cashflow	4,094	-1,682	-4,496	-6,684	-8,600
Cashflow from operations	4,784	5,258	12,970	19,211	33,883
Capital expenditure	-2,005	-2,301	-3,727	-5,998	-7,490
Free cashflow	2,779	2,957	9,243	13,213	26,393
Reduction in investments	-340	-251	-1,701	0	0
Net acquisitions	0	0	0	0	0
Dec in other LT assets	0	0	0	0	0
Inc in other LT liabilities	0	0	0	0	0
Adjustments	454	3,980	255	0	0
CF after investing acts	2,893	6,686	7,797	13,213	26,393
Cash dividends	-2,897	-3,905	-8,292	-12,930	-19,445
Equity issue	0	0	0	0	0
Debt issue	393	-200	35	0	0
Convertible debt issue	0	0	0	0	0
Others	-422	-459	-35	0	0
CF from financial acts	-2,926	-4,563	-8,292	-12,930	-19,445
Net cashflow	-33	2,123	-496	283	6,949
Beginning cash	4,132	4,099	6,222	5,727	6,009
Ending cash	4,099	6,222	5,727	6,009	12,958
Ending net debt	-574	-2,904	-2,372	-2,655	-9,603

Balance sheet (TWDmn)

As at 31 Dec	FY24	FY25	FY26F	FY27F	FY28F
Cash & equivalents	4,099	6,222	5,727	6,009	12,958
Marketable securities	941	2,092	3,747	3,747	3,747
Accounts receivable	6,070	8,925	16,073	23,957	26,604
Inventories	5,458	7,919	10,704	15,732	17,003
Other current assets	856	967	1,351	1,351	1,351
Total current assets	17,425	26,126	37,601	50,797	61,662
LT investments	8,918	7,953	8,791	9,814	10,956
Fixed assets	6,956	7,232	10,059	14,659	20,144
Goodwill	0	0	0	0	0
Other intangible assets	0	0	0	0	0
Other LT assets	4,009	5,839	5,797	5,797	5,797
Total assets	37,308	47,150	62,249	81,067	98,560
Short-term debt	1,417	14	305	305	305
Accounts payable	3,106	4,116	6,190	9,097	9,832
Other current liabilities	3,711	5,030	7,308	8,130	8,406
Total current liabilities	8,235	9,159	13,802	17,532	18,543
Long-term debt	2,108	3,304	3,050	3,050	3,050
Convertible debt	0	0	0	0	0
Other LT liabilities	1,515	2,125	2,082	2,082	2,082
Total liabilities	11,858	14,587	18,935	22,665	23,676
Minority interest	573	665	972	1,212	1,452
Preferred stock	0	0	0	0	0
Common stock	8,851	8,465	8,518	8,518	8,518
Retained earnings	15,163	22,825	32,990	47,838	64,081
Proposed dividends	0	0	0	0	0
Other equity and reserves	863	608	833	833	833
Total shareholders' equity	24,877	31,898	42,341	57,189	73,432
Total equity & liabilities	37,308	47,150	62,249	81,067	98,560

Liquidity (x)

Current ratio	2.12	2.85	2.72	2.90	3.33
Interest cover	-	-	-	-	-

Leverage

Net debt/EBITDA (x)	net cash	net cash	net cash	net cash	net cash
Net debt/equity (%)	net cash	net cash	net cash	net cash	net cash

Per share

Reported EPS (TWD)	12.49	27.70	43.59	65.56	84.23
Norm EPS (TWD)	12.49	27.70	43.59	65.56	84.23
FD norm EPS (TWD)	12.44	27.59	43.42	65.30	83.90
BVPS (TWD)	58.72	75.30	99.96	135.01	173.36
DPS (TWD)	9.04	19.58	30.52	45.90	58.97

Activity (days)

Days receivable	102.6	96.7	85.3	97.4	98.8
Days inventory	224.9	224.3	169.1	173.5	176.2
Days payable	128.0	121.1	93.6	100.3	101.9
Cash cycle	199.5	199.9	160.8	170.6	173.1

Source: Company data, Nomura estimates

Company profile

Chroma is a world leading supplier of precision test and measurement instrumentation, automated test systems, intelligent manufacturing systems, and test & automation turnkey solutions.

Valuation Methodology

Our TP of TWD2,845 is based on 38x average 2027-28F EPS, at the higher end of Chroma's historical trading band (14-58x). The benchmark of this stock is TAIEX.

Risks that may impede the achievement of the target price

Major downside risks to our call include: 1) CoWoS and backend testing capacity expansion slowdown; 2) slower-than-expected product refresh, platform performance upgrade, or ramp-up by the AI chip end customers; 3) fiercer-than-expected from other international testing equipment makers; and 4) weaker-than-expected end-market demand, particularly for AI servers.

ESG

The company is fully aware of climate change and the need for carbon reduction. It is dedicated to minimize the use of energy and resources as well as reducing the amount of waste produced in manufacturing processes. At the same time, Chroma has been continuously committed to social welfare and care for underprivileged groups and young children.

Semiconductor/Photonic Test – AI/HPC and CPO pave growth trajectory

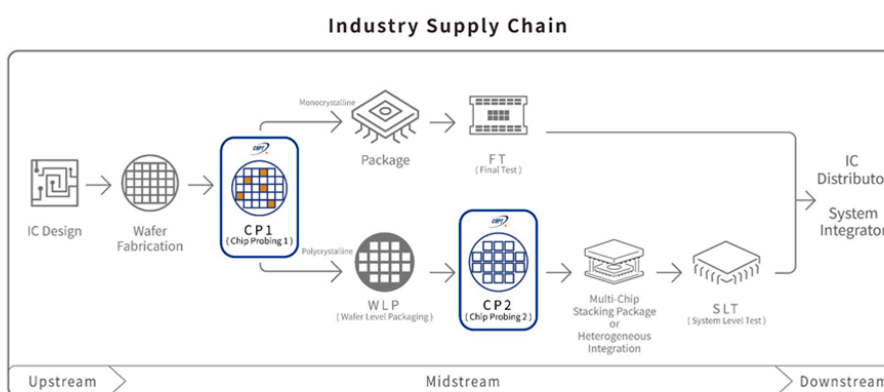
We estimate Chroma's Semiconductor/Photonic Test business to register a 68% revenue CAGR over 2025-28F, riding on the secular semiconductor testing tailwinds from increasing design complexity and larger footprints particularly in AI/HPC chips, as well as the emergence of CPO. Chroma derives 38% of its revenue from Semiconductor/Photonic Test in 2026F, in which we estimate a 70-80% and 20-30% split between semiconductor and photonics, with the company mainly providing test handlers for SLT used in niche, quality-dependent verticals like AI/HPC.

We believe Chroma's SLT handlers will enjoy an intense equipment upgrade cycle given regular AI/HPC platform refresh cadences, and will particularly capitalize on the ASP uptrend from **nVidia's AI GPU migration to SoIC stacking** in 2028E for more stringent thermal control requirements (we estimate nVidia currently accounts for ~20% of Chroma's total revenue). TSMC and OSATs' more aggressive **2.5D/CoWoS capacity investments** and **more testing spending by AI OSATs** could be positive indicators for Chroma's SLT handler business, and we see potentially more SLT adoption by AI/HPC chips in the longer run as well, contemplating more unbearable field return costs. Thus far, Chroma has secured new SLT opportunities from **Google** (GOOGL US, Not rated) **Axion CPU** and **AMD** (AMD US, Not rated) **MI450 series**. We also like Chroma's progress into **2D/3D wafer metrology** to capture another adjacency of advanced packaging capacity growth. Despite no uniform testing standard for CPO and uncertainty about ramp-up timeline, we expect Chroma's established experience in photonics to pave the way for its handler role in **test insertion 3** and optical tester in **test insertion 4**.

FT and SLT are Chroma's testing equipment addressable markets

The semiconductor backend processes are punctuated by a few test insertions that sort out defective silicon as early as possible to gate output quality – **chip probe (CP)**, **final test (FT)**, **burn-in test (BIT)**, and **system-level test (SLT)**, and the entire backend flows start with CP at the wafer level before the individual dies are cut out of the wafer and undergo subsequent assembly. We explore the IC backend testing process in our [Anchor Report](#), and here we only focus on FT and SLT, which are Chroma's addressable segments.

Fig. 1: An illustration of semiconductor manufacturing and backend testing flows



Source: CHPT, Nomura research

FT is a mandatory procedure, and depending on IC customers' configurations, there could be two or more FT steps during the manufacturing cycle (e.g., nVidia's AI GPUs require two FT steps, one before BIT and the other after BIT). FT is carried out after IC assembly (on substrates, lead frames, or wire bond) and chip packages are capable of accommodating larger input voltages or electric current than bare dies. FT measures electrical properties of chips given pre-programmed inputs to identify product functionality ("pass or fail" is the terminology) and sort functional products by grade (also known as

“binning”), **all within the shortest possible time frame (usually hundreds or thousands of seconds)**, as chip vendors may want to control test charges (measured by “hourly rates”) which are amortized expenses of costly testers.

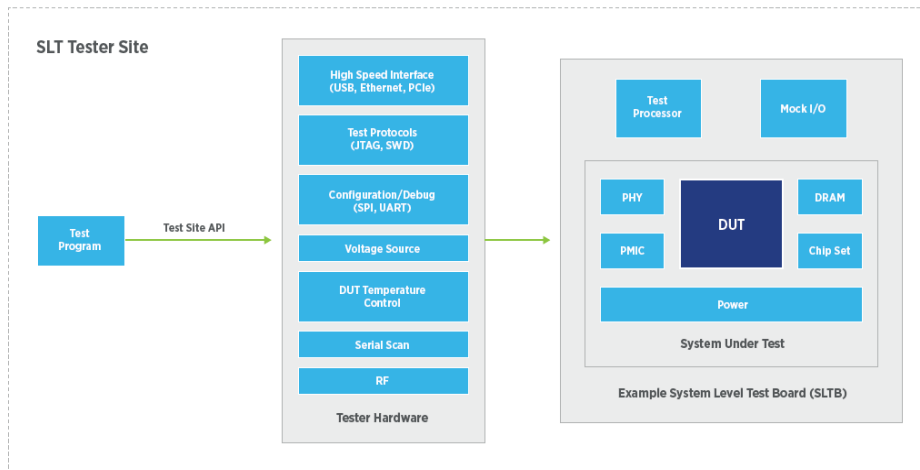
We discuss digital IC testing in detail in this report. We note below three primary parameters that form a sequential filtering system in FT for logic circuits whose focuses are on whether outputs are “correct”. By contrast, analog/mixed-signal IC testing places more emphasis on the “measurement accuracy” of select parameters (e.g., linearity and signal-to-noise ratio).

1. **Direct current (DC) test:** The DC test is the first phase in FT and consumes the least time amongst the three phases. It measures the steady-state electrical characteristics of the chip using direct current and ensures the physical silicon structure is intact. The DC test does not validate logic or speed, however. The primary metrics measured during this phase include: 1) **open/short test** – which verifies all pins are properly connected to the internal circuitry (no “open”) and that no pins are accidentally welded together (no “short”); 2) **leakage current** – which ensures a pin does not “bleed” excessive current into the substrate when it is given a specific voltage level; 3) **power consumption** – which detects the electric current drain inefficiency; and 4) **output voltage level** – which confirms the chip can drive signals out at the correct voltage thresholds required to communicate with other components.
2. **Function test:** The subsequent function test is to check whether the logic operation works properly. During the function test, the tester floods the chip with pre-programmed binary inputs (i.e., test vectors) and records the outputs generated by the chip to compare the results against the expected mathematical truth table. Function tests are typically executed at a nominal, conservative clock speed to isolate pure logical errors from high-speed timing anomalies.
3. **Alternating current (AC) test:** The AC test introduces the critical dimension of “time”, evaluating how the chip performs under dynamic, high-frequency AC conditions and checking the output signals’ “waveform”, because a chip might have perfect structure and flawless logic responses at slow speeds. The AC test pushes the silicon to its physical limits by measuring sub-nanosecond timing parameters to ensure signals propagate cleanly across the die without corruption. Major test items include propagation delay, setup and hold times, rise and fall times, and maximum frequency. The AC test is also **the foundation of “binning”**, sorting out the best-performing chips and lower-tiered ones.

On the other hand, **SLT is an optional procedure**, during which the chip is inserted into a system test board designated by the chip vendor to simulate how it works in the real-world scenario. The test board is a modified version of the actual commercial motherboard that eventually houses the silicon and is equipped with other components such as memory and peripherals. The SLT boots a full operating system like Linux or Android and runs heavy, asynchronous software workloads to replicate genuine user environments. Throughout the process, SLT measures functional and behavioral metrics rather than raw structural data, including system stability, workload execution throughput, thermal throttling thresholds, and high-speed interface bit error rates (BER).

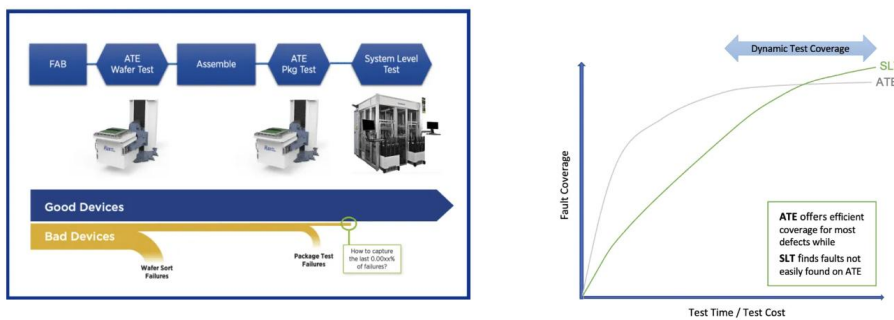
SLT has gained traction industry-wide because it exposes defect classes that the preceding stages structurally cannot. DC test, Function test, and AC test are largely deterministic and exercise the device in isolation, at fixed patterns and controlled conditions. They are effective at catching hard, static failures but are poorly suited to intermittent, workload-dependent or interaction-level defects – issues that only manifest when multiple IP blocks operate concurrently under realistic power, thermal, and timing stress, or under specific software-induced corner cases. As chip complexity has grown with heterogeneous integration, multi-core coherency, and higher-speed interfaces, these system-level **“test escapes”** could result in a larger share of field returns, which has steered SLT from a niche practice in automotive/mobile applications toward broader adoption across compute, AI accelerator, and networking chips, where the cost of a return merchandise authorization (RMA) far outweighs the cost of additional test screening.

Fig. 2: SLT acknowledges that software is part of the system, and recreates the end-use environment as closely as possible



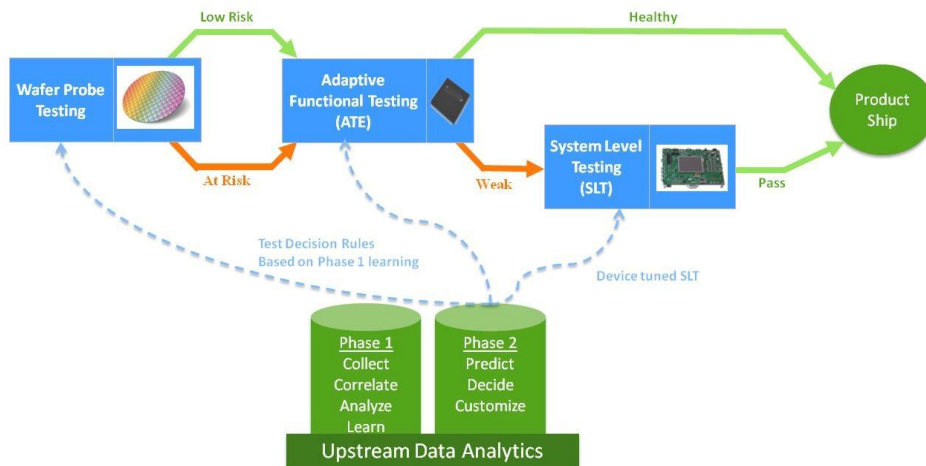
Source: Teradyne, Nomura research

Fig. 3: Traditional test coverage becomes more challenging, and SLT's strategic importance is emerging



Source: Teradyne, Nomura research

The SLT phase is less expensive than FT given there is no usage of multi-million-dollar testers, but it is **very time-consuming** and could take minutes or even hours to execute per devices. At high volumes, the throughput mismatch could become the binding constraint on SLT adoption, since running every unit through hours of system-level workloads is economically impractical for most product lines. As such, the industry leans toward parallel testing inside automated SLT handlers, and potentially the integration of adaptive testing backed by machine learning algorithm which selectively routes only silicon at risk to extended SLT screening while the bulk of the chips proceed through a shortened or standard flow. This "smart SLT" approach preserves most of the defect-screening benefit while containing the test time and cost overhead, and is emerging as a key differentiator among OSATs and IDMs competing on both quality (defined by defective parts per million, or DPPM) and test cost per unit.

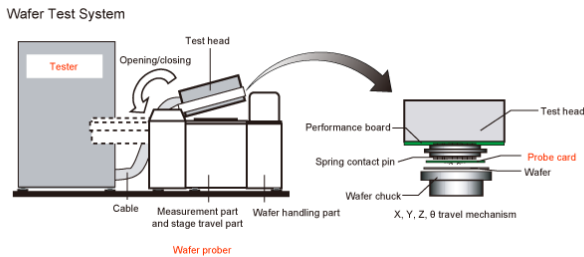
Fig. 4: Adaptive SLT could further reduce the overall cost (time) of test

Source: AEM, Nomura research

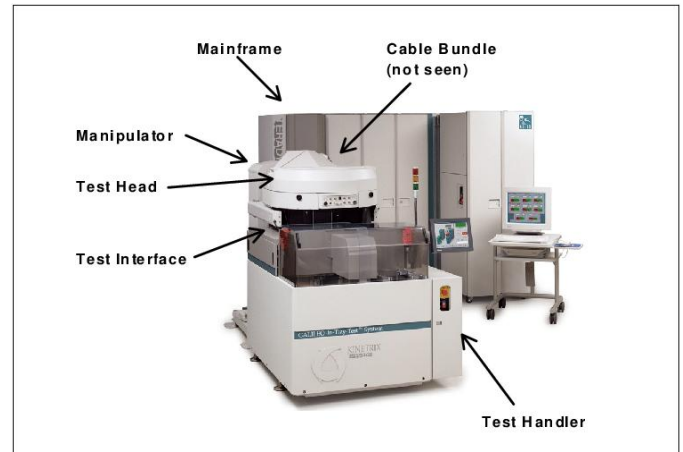
IC test trios: ATE, prober/handler, and test interface

The semiconductor testing is not carried out by one single tool but a tightly integrated system also known as “test cell”. The test cell consists of three major parts: automated test equipment (ATE), prober/handler, and application-specific test interfaces. In SLT, the ATE is replaced with a system test board on which test sockets are equipped, housed inside an SLT handler.

- **Automated test equipment (ATE):** ATE, or simply “tester”, is the sophisticated core workstation of the test cell that feeds precise **electrical stimulus signals** (e.g., voltages, currents, or high-frequency waveforms) into the device under test (DUT) and monitors the output responses. The tester compares the chip’s real-time performance running on pre-programmed inputs against engineering design specifications to identify defectives. In the ATE architecture, the mainframe houses the tester’s power supply, central cooling unit and the system controllers, and the test head houses the test interfaces.
- **Prober/handler:** The prober or the handler is the mechanical automation arm of the test cell that operates with the ATE to ensure a continuous, high-speed, and unmanned silicon test flows. The test head is flipped and pneumatically or hydraulically docked into the core mechanical nest of the prober/handler. The prober and the handler are used in different testing scenario – **the prober is present in CP and handles uncut wafers**, using microscopic pins to physically contact with individual die before they are singulated, while **the handler picks up chip packages (in FT or SLT)** from trays, inserts them into the test sockets, and **physically sorts them into different bins** based on the tester’s verdict.
- **Test interface:** The test interface is the physical and electrical “bridge”, customized based on the silicon layout/package, to connect the generic tester to the DUT. A probe card paired with pogo pins (or MEMS pins for fine-pitch applications) is being used in CP, while a load board with test sockets is adopted in FT.

Fig. 5: A test cell in chip probe

Source: MJC, Nomura research

Fig. 6: A test cell in final test

Source: SAE Technical Papers "Thermal Management and Control in Testing Packaged Integrated Circuit (IC) Devices", Nomura research

The IC test squad is not complete without “role players”

Beside the aforementioned trios, we note there are also tools supplementing temperature management to prevent undesired shifts in chip performance and physical properties under different thermal conditions, and specialized plug-in modules to expand or reconfigure the tester's scope.

- **Thermal chuck:** During CP, the chuck is the flat, rigid metallic platform that holds the silicon wafer in place using vacuum suction. A thermal chuck is a chuck with built-in heating elements and internal cooling channels to uniformly control the temperature of the wafer under test when mimicking different thermal conditions in operations. A thermal chuck must demonstrate extreme planarity (i.e., very meager warpage) at extreme temperature swings to avoid poor electrical contacts or wafer crack by probe needles.
- **Active thermal control system (ATC):** ATC is a dynamic temperature management system integrated into the handler. When a tester boots up workloads, the DUT could undergo a spike in internal power density and instantaneously heat up. ATC detects the thermal conditions of the chip under test and instantly adjusts its cooling/heating action to counteract the chip's internal power fluctuations.
- **Instruments:** The tester is essentially a modular chassis populated by a cluster of instruments that define its initial testing scope. However, if a silicon requires test coverage beyond this built-in set, specialized instruments can be added to the tester to extend its capabilities. These modular instruments are notably critical for high-performance domains like analog/mixed-signal and high-speed digital, and we observe optical instruments are introduced in CPO test insertions to assist with photonic test items (e.g., optical insertion loss, spectrum/wavelength and polarization effects).

Fig. 7: The thermal chuck is integrated into the wafer prober

Thermal chuck (left; by ATT Systems) and wafer prober (right; by FormFactor)



Source: FormFactor, Nomura research

Fig. 8: The ATC is integrated into the handler

Source: Hon Precision, Nomura research

Fig. 9: The instrument helps expand dedicated test coverage

Source: Keysight, Nomura research

Fig. 10: Semiconductor test equipment and interface maker overview

Tester/ATE	Wafer prober	Chip probe		Test interface (probe card/probes)	Test interface (PCB/substrate only)	Handler Active thermal control	Final test/System level test		Instrument
		Wafer chuck	Thermal chuck				Test interface (socket)	Test interface (load board)	
Advantest (6857 JP)	Tokyo Electron (8035 JP)	ERS Electronics (unlisted; working with MPI)		*Technoprobe (TPRO IM)	Daeduck (353200 KS)	Hon Precision (7769 TT)	Winway (6515 TT)	Advantest (6857 JP)	Keysight (KEYS US)
Teradyne (TER US)	Tokyo Seimitsu (7729 JP)	ATT Systems (unlisted; working with FormFactor)		*FormFactor (FORM US)	Gorilla Circuits (unlisted)	Kanematsu (8020 JP)	Yamaichi (6941 JP)	Technoprobe (TPRO IM)	Rohde & Schwarz (unlisted)
Cohu (COHU US)	Advantest (6857 JP)	inTEST (INTT US)		*CHPT (6510 TT)	Fastprint (002436 CH)	Cohu (COHU US)	LEENO (058470 KS)	CHPT (6510 TT)	Tektronix (Ralliant [RAL US])
SPEA (unlisted)	Techwing (089030 KS)			*MPI (6223 TT)		AEM (AEM SP)	Enplas (6961 JP)	KSMT (6683 TT)	National Instruments (Emerson [EMR US])
Chroma (2360 TT)	FormFactor (FORM US)			*Cohu (COHU US)		TESEC (6337 JP)	Yokowo (6800 JP)	TSE (131290 KS)	Anritsu (6754 JP)
TESEC (6337 JP)	MPI (6223 TT)			*TSE (131290 KS)		SRM (unlisted)	Smiths/DI (SMIN LN)	Daeduck (353200 KS)	Viavi (VIAV US)
Accotest (688200 CH)	SEMICS (unlisted)			*ZENFOCUS (unlisted)		Techwing (089030 KS)	ISC (095340 KS)	Gorilla Circuits (unlisted)	
Changchuan (300604 CH)	Powertech (301369 CH)			*Maxone (688809 CH)		Chroma (2360 TT)	Cohu (COHU US)	Fastprint (002436 CH)	
Powertech (301369 CH)				*JEM (6855 JP)		Advantest (6857 JP)	TSE (131290 KS)	ZENFOCUS (unlisted)	
				*MJC (6871 JP)		SEMES (unlisted)	MJC (6871 JP)		
				Winway (6515 TT)		Changchuan (300604 CH)			
				KSMT (6683 TT)		JHT (603061 CH)			
				*Has in-house pins		Powertech (301369 CH)			

Source: Company data, Nomura research

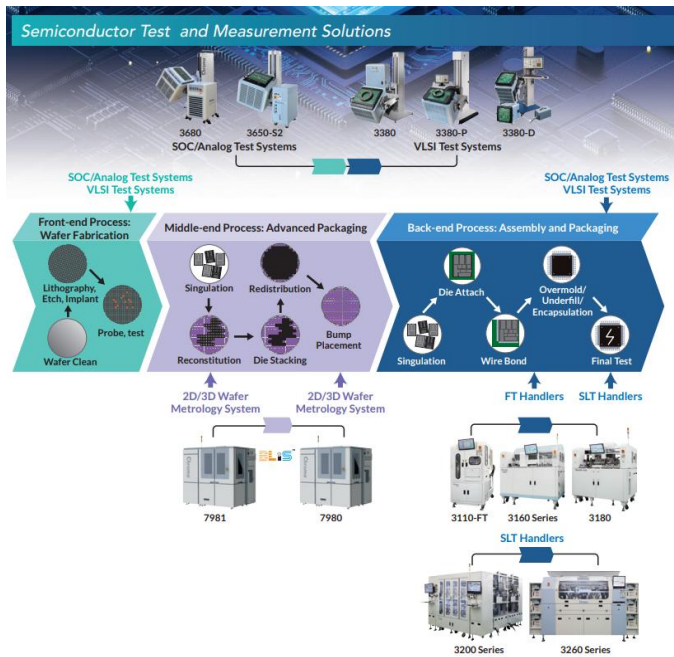
Chroma is an expert in niche testers and ATC/handlers

Chroma has strategically positioned itself in the semiconductor backend testing arena with a focus on niche applications, rather than chasing the broad-market, leading-edge ATE segment dominated by the largest global players such as Advantest (6857 JP, Buy) and Teradyne (TER US, Not rated). Chroma's focus is most visible in ATC (named **"temperature forcing systems"** at Chroma) and /handlers (primarily pick-and-place handlers, which uses suction to transfer the DUT and press it into the test sockets) and SoC testers for mature logic or analog/mixed-signal semi.

In ATC/handlers, the company specializes in FT/SLT (**with more focus on SLT**) and offers both **dual-temp** (high temperature [up to 150°C] and ambient temperature [around 25°C]) and **tri-temp** (high temperature, ambient temperature, and low temperature [down to -70°C]) ATC capabilities. Chroma's handlers add multi-site flexibility across single, dual, quad, octa-site configurations or more. **AI/HPC applications**, for instance, is a Chroma's target vertical for SLT handlers, and we believe nVidia is a key customer of Chroma.

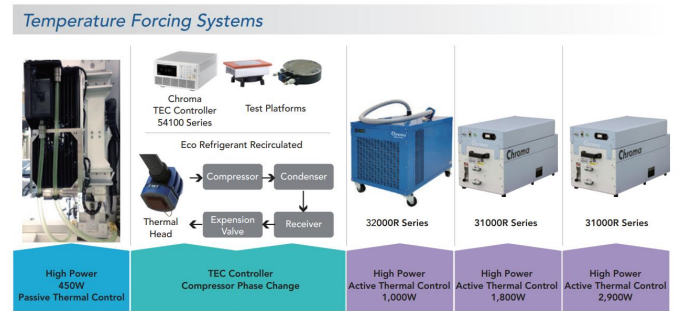
On the SoC tester side, Chroma's niche is **mature-node logic or analog/mixed-signal semi** such as microcontroller units (MCUs) and power management ICs (PMICs), supporting global fabless and IDM customers, as well as Chinese ones, rather than competing head to head with global leading players in the advanced logic test.

Fig. 11: Chroma's semiconductor test solutions



Source: Company data, Nomura research

Fig. 12: Chroma's temperature forcing system can be seamlessly integrated into Chroma's handlers and optimized for mass parallelism



Source: Company data, Nomura research

Taiwan peer comparison – Chroma has little overlap with Hon. Precision

A frequently asked question by investors is whether and where Chroma competes with another Taiwan-based backend testing equipment maker Hon. Precision (7769 TT, Buy). Admittedly, both the companies have ATC/handler offerings for FT and SLT, but we believe they have fairly little business overlap given Chroma's focus on SLT handlers (we estimate 60-70% of Chroma's semiconductor testing revenue comes from SLT) vs Hon's focus on FT handlers (70-80% of handler revenue comes from FT).

Hon has been making inroads into the SLT handler market by leveraging its competency in customization and ATC, and we believe AMD is one of its longstanding customers in SLT handlers, although we think Chroma could have secured SLT handlers for AMD's latest MI400-series. We are aware that Hon has started engineering collaboration with nVidia in SLT handlers for the future generation of AI GPUs, but we are uncertain about whether Hon could really challenge Chroma's dominance in this segment. In our view, Hon's new SLT handler opportunities could lie squarely within ASIC customers whose broader adoptions of SLT, apart from mandatory FT, have significantly increased.

Fig. 13: Comparison of major global tester and handler makers

USD mn	Headquarter	Market cap	2025 revenue	Revenue mix		2025 GM	2025 OPM	2025 net profit	Tester	Handler
				Tester	Handler					
Chroma ATE (2360 TT)	Taiwan	24,332.8	908.4	10-15%	20-25%	61.5%	32.5%	375.2	Mature logic/analog	FT 30-40%, SLT 60-70% Mostly logic/analog
Advantest (6857 JP)	Japan	123,945.2	6,902.1	82%	<5%	62.4%	39.7%	1,927.8	80% Logic/Analog 20% Memory	FT, SLT Mostly memory
Teradyne (TER US)	US	50,462.1	3,190.0	75%	15%	58.3%	22.3%	632.1	80% Logic/Analog 20% Memory	SLT Mostly logic/analog
Cohu (COHU US)	US	2,414.9	453.0	~40%		43.3%	0.5%	(10.1)	Mature logic/analog	FT, SLT Mostly logic/analog
Hon. Precision (7769 TT)	Taiwan	33,781.0	971.3	n.a.	c.80%	56.5%	49.7%	396.6	n.a.	FT 70-80%, SLT 20-30% Mostly logic/analog
Kanematsu (8020 JP)	Japan	2,157.8	7,062.3	n.a.	<5%	15.7%	4.4%	215.8	n.a.	FT, SLT (from Seiko Epson) Mostly logic/analog
TESEC (6337 JP)	Japan	75.5	37.6	45%	36%	37.8%	5.7%	2.0	Mature logic/analog Discrete	FT (from Yokogawa) Mostly logic/analog
Techwing (089030 KS)	South Korea	1,222.7	112.0	n.a.	36%	42.2%	10.0%	6.6	n.a.	FT Mostly memory
SEMES (unlisted)	South Korea	n.a.	n.a.	n.a.		n.a.	n.a.	n.a.	n.a.	FT Mostly memory
AEM (AEM SP)	Singapore	2,137.2	305.6	30%		25.7%	5.9%	13.0	Mature logic/analog	FT, SLT Mostly logic
Changchuan (300604 CH)	China	26,090.7	736.2	c.60%	c.30%	55.1%	24.4%	185.2	Logic/Analog Discrete	FT, SLT Logic/Analog, Discrete
Accotest (688200 CH)	China	11,034.7	187.3	c.90%	n.a.	73.8%	36.5%	74.6	Logic/Analog Discrete	n.a.
JHT (603061 CH)	China	3,917.9	97.1	n.a.	c.90%	52.2%	30.5%	24.6	n.a.	FT, SLT Logic/Analog, Discrete
Powertech (301369 CH)	China	2,233.8	49.3	c.85%		55.7%	-3.1%	4.7	Logic/Analog Discrete	FT Logic/Analog, Discrete

Note: Market cap data as of July 17, 2026

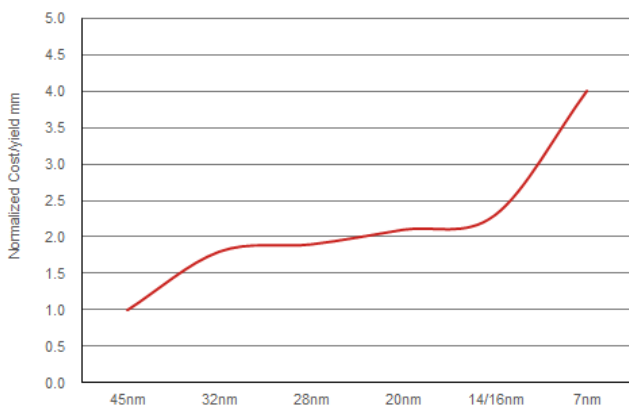
Source: Company data, Bloomberg Finance LP, Nomura estimates

Increasing design complexity and larger footprints of AI/HPC chips spur structural testing growth

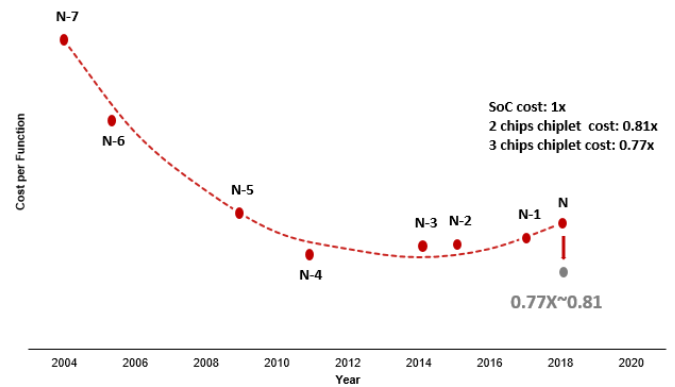
We believe **increasing chip design complexity** and **larger package footprints** are the two most critical catalysts inside AI/HPC chips that are fostering structural growth in the semiconductor testing process, an often overlooked element in the manufacturing value chain because of its relatively low portion in the cost structure. The semiconductor industry is currently undergoing a paradigm shift in layout/architecture by embracing a chiplet-based design philosophy, 2.5D advanced packaging to stitch multiple-reticle-sized interposers and 3D IC stacking to increase performance per area (measured by transistor density), as conventional geometric scaling slows, costs for chip design and yielding large die continue to rise at more advanced logic nodes (*Fig. 14*, *Fig. 15*, and *Fig. 16*), and any single chip inevitably faces the physical limit of reticle size (26x33mm exposure field size, or 858mm²). In a seminar during SEMICON Taiwan 2025, TSMC estimated compute performance per reticle improvement by 80x from N28 to A16, and the incorporation of advanced packaging could enlarge the gain to ~320x at >9.5x-reticle CoWoS (*Fig. 17*).

Fig. 14: The cost per yielded die vs node migration

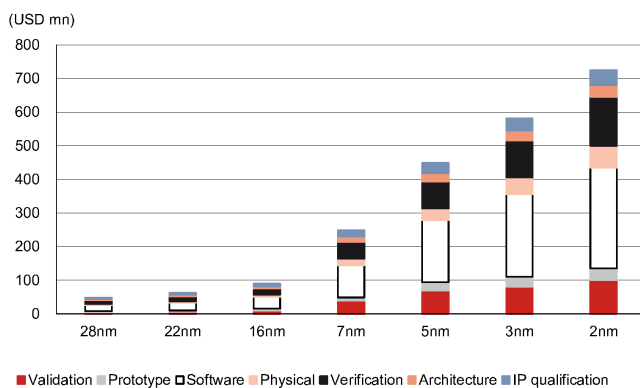
The cost per yielded die continue to increase when entering into more advanced nodes



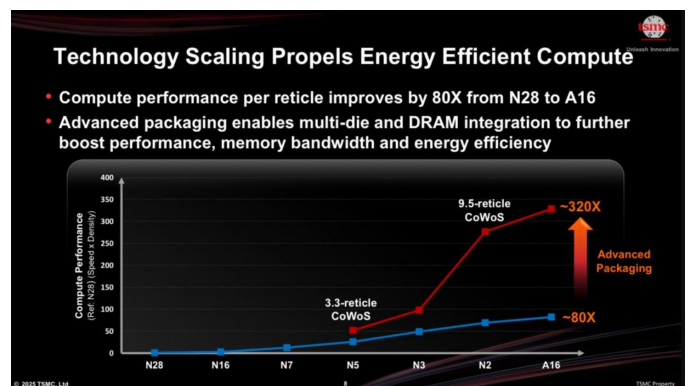
Source: Nomura research

Fig. 15: Cost reduction by using chiplets vs SoC solution

Source: TSMC, Nomura research

Fig. 16: Skyrocketing chip design cost moving to more advanced nodes

Source: IBS, Nomura research

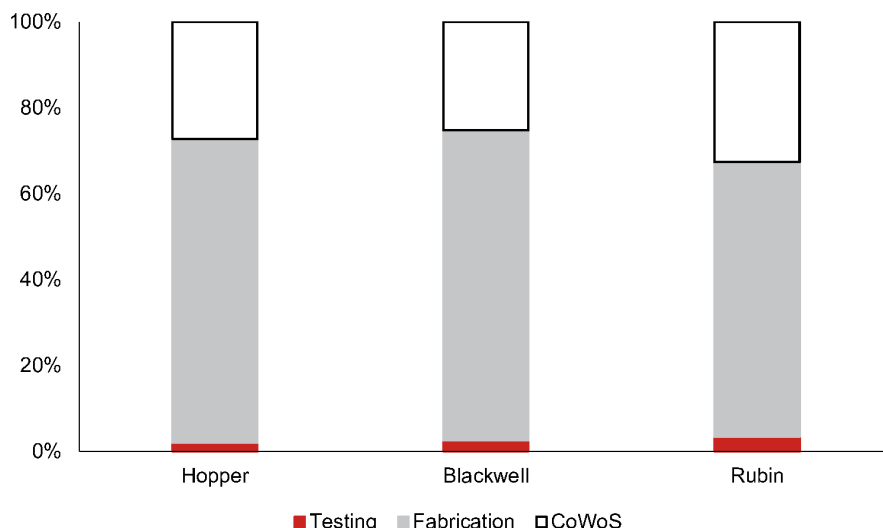
Fig. 17: Advanced packaging is an approach to bring the subsystem performance further up

Source: TSMC, Nomura research

While these innovations enable unprecedented computing power for AI/HPC, they have fundamentally altered the economics and mechanics of semiconductor manufacturing.

Die-to-die interconnect in chiplets or 2.5D/3D packaging is a complex jigsaw puzzle replacing traditional seamless on-die communication, because engineers must ensure robust **signal and power integrity**, and **thermal management** and **mechanical stress control** also become crucial in such layouts. Consequently, we believe semiconductor testing is no longer just a routine quality-control process at the end of the production flows, and has become an indispensable, high-stakes discipline that directly dictates commercial success.

As device complexity increases, a longer testing time is needed, and if we take nVidia AI GPUs as examples and index the SLT time of Hopper to 1, we reach 1.5x for Blackwell's SLT time and 2.5x for Rubin's. In our view, the prolonged testing time and higher hourly rate should add to the testing content under cost. We estimate the overall testing content (defined as the sum of FT, BIT and SLT) in nVidia Rubin could rise to 3.3% of total cost vs. 2.5% for Blackwell and 1.9% for Hopper, based on our supply chain checks and analysis (Fig. 18). The lengthened testing time presents an outright tailwind for semiconductor testing equipment makers, in our view, given the need by OSATs to keep up with throughputs.

Fig. 18: Increasing testing content in the logic manufacturing cost of nVidia AI GPUs

Source: Company data, Nomura estimates

More intense upgrade cycles for SLT handlers than FT handlers

We believe SLT handler upgrades are primarily driven by the tight coupling between the handler's mechanical and thermal interface and the physical design of each new AI chip platform. Unlike FT, which tests a packaged die in a relatively standardized form factor, SLT validates the chip **at the full system or module level**, indicating socket pin count, board layout, power delivery, and thermal load are all specific to that generation's module configurations.

Socket pin count is a key driver as each new AI/HPC chip tends to add more pins to support more power rails and higher I/O bandwidth. A pin-count change would therefore alter pin pitch, contact force distribution, and co-planarity tolerances, and the SLT socket typically must undergo redesign and re-qualification efforts when the AI chip moves to a new generation. The **system test board layout** compounds the changes, since system test boards are built to mirror the actual system operations including the power delivery network and high-speed signal topology, and the change in the system architecture could be very significant (e.g., from OAM+UBB in Hopper to Bianca in Blackwell).

Meanwhile, as AI/HPC silicon scale **power density** along with platform refresh cadence (even at an accelerated pace), the **thermal subsystem** has to scale in lockstep. Reliability and thermal control requirements during a test could also foster the upgrade vector. All in all, generation-specific parameters of AI/HPC chips necessitate test interface re-engineering, as well as compress the effective upgrade cycle and foster OSATs' purchases of new SLT handler platform at a higher ASP.

Compared to FT handlers, the generation-specific variables lead to a more frequent upgrade of SLT handlers. FT handlers could be insulated from full replacement as the core pick-and-place mechanics like trays, robotic arms and contact heads may persist across multiple device generations with the ATC and the mechanical conversion interface requiring regular updates. Although OSATs are adding ATC/handler capacity to address surging AI demand, we believe most of the upgrades (and therefore ASP growth) are within the ATC and with nearly no change in the handler – unless there is mechanical backbone redesign for larger trays or completely new device form factors (e.g., microchannel lids [MCL]).

More stringent thermal control requirements for SLT, moving toward SolC

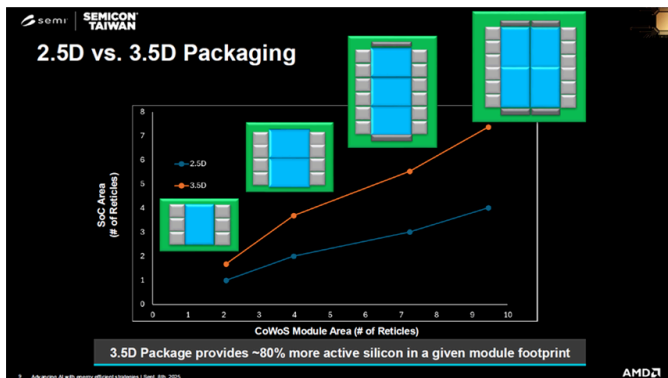
We believe Chroma will particularly capitalize on the ASP uptrend from nVidia's AI GPU migration to SolC stacking, since the sub-system performance gain from more dies in a package comes at the cost of more stringent thermal requirements.

The max TDP of nVidia Blackwell Ultra (B300) is 1,400W, and we think the TDP of Rubin (R100) may start from 1,800W. We foresee further elevating thermal challenges moving to Feynman (F100), slated for 2028E (Fig. 23). nVidia at its GTC 2026 unveiled a plan to

adopt 3D stacking starting from the Feynman platform (*report*). Vertical chip stacking (SoIC platform at TSMC) theoretically could augment transistor counts per package, an outright indicator of computing power, without extra footprints (vs CoWoS/2.5D packaging that expands horizontally to accommodate more chips). AMD (AMD US, Not rated) believes 3.5D chip modules (i.e. 2.5D packaging + 3D hybrid bond) could enable shorter data paths and improved interconnect energy efficiency, and estimates ~80% more active silicon in a given module footprint than sheer 2.5D packaging (*Fig. 19*). We think the 3D logic stacking approach may also lie in other custom AI chip future roadmap (*Fig. 20*).

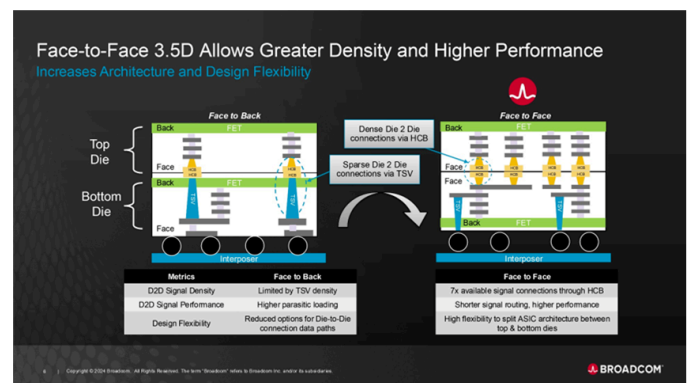
Currently, AMD leads the adoption of SoIC at TSMC (starting from MI300-series), and in the latest MI450, we believe AMD stacks four top dies (four XCD) on two reticle-sized active interposers (I/O dies), in which each top die scales to about 1/3 reticle size (*Fig. 21*). We observe nVidia will be more aggressive in chip specs by stacking a reticle-sized GPU die on top of another for the Feynman platform, the first-ever GPU-on-GPU SoIC stacking, which would lead to higher computational power even with limited growth in interposer reticle stitching size (c.6x reticle, see footprint in *Fig. 22*; up from c.5x in Rubin). Such a practice theoretically exacerbates thermal dissipation challenges, and we expect it could spark the adoption of **more powerful ATC along with Chroma's SLT handler at a higher ASP**.

Fig. 19: 3.5D packaging offers a more dense chiplet module



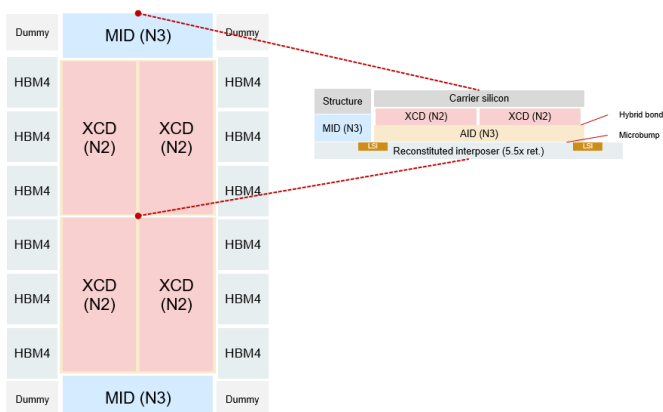
Source: AMD, Nomura research

Fig. 20: Broadcom introduced face-to-face 3D hybrid bond on its 3.5D platform



Source: Broadcom, Nomura research

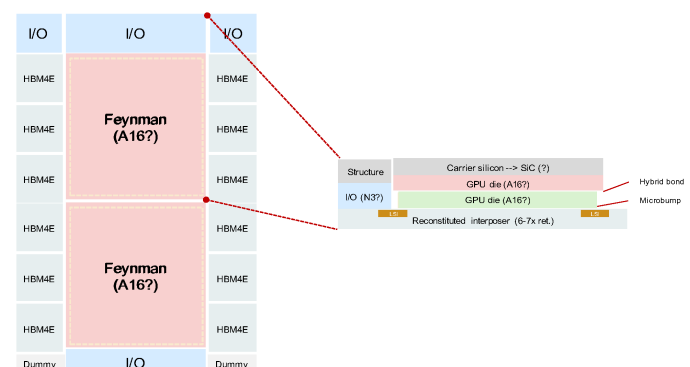
Fig. 21: Floorplan of AMD MI455 and cross section



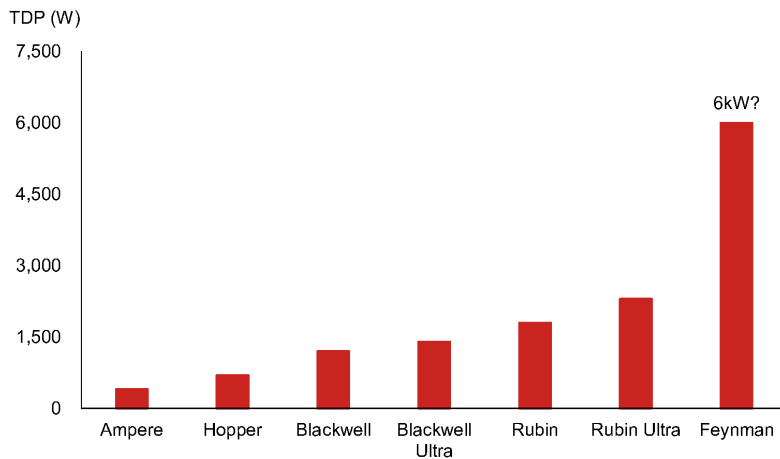
Source: Company data, Nomura research

Fig. 22: The floor plan and cross-section chart of nVidia's Feynman GPU

SiC thermal plate to function as an integrated silicon carrier (fill up the height gap in between GPU and HBM) and thermal interface material (TIM)



Source: Company data, Nomura estimates

Fig. 23: TDP of nVidia products

Source: Company data, Nomura estimates

Continued CoWoS capacity growth at TSMC, fueled by AI/HPC

We think TSMC's CoWoS capacity growth is **one proxy (albeit imperfect)** for Chroma's SLT handler sales volume trend as both are primarily fueled by AI/HPC demand, and Chroma's key SLT handler customers nVidia and AMD (starting from MI400-series) together could consume about 65% of CoWoS capacity each year, as per our estimates. We also estimate that c.70% of Chroma's Semiconductor/Photonic test revenue comes from AI/HPC in 2026F. But, we note that there is not necessarily a mathematical relationship between CoWoS capacity builds and handler purchases given varying output per interposer wafer for different AI accelerators and the throughput factor in handlers.

We do admit that one caveat of using the CoWoS capacity growth to sketch the growth trajectory of Chroma's AI SLT handler business is that SLT is not yet a mandatory stage during the semiconductor backend process. However, traditional ATE-driven FT alone may be no longer sufficient to detect all structural and functional defects since logic chip packages become more complicated, and from a structural standpoint, we foresee more ubiquitous adoption of SLT going forward, particularly for AI chips whose field returns could be extremely costly. These secular tailwinds are clearly beneficial for leading AI SLT handlers like Chroma, in our view.

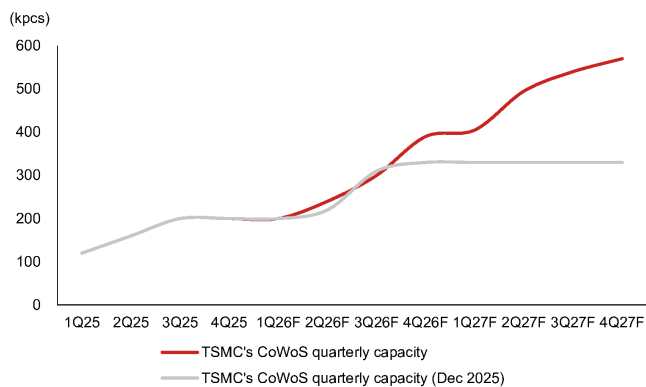
To date, TSMC's supply is apparently still constrained across the front-end and the back-end given the demand strength from AI, and the company has expressed its commitment to expand its capacity in due course, citing that it *"works very hard to meet all the demand"* and *"doesn't leave any business on the table"* (see [remarks from 4Q25](#) and [1Q26 results](#)). We observe TSMC has turned more aggressive on CoWoS capacity planning (or more precisely, "CoW" capacity) in recent months in response to surging AI chip demand, and lifted the capex guidance again in the July earnings call ([report](#)), which should indirectly underscore the growth trajectory of Chroma's SLT handler business.

In our most recent [Asia AI Semi & Server Anchor Report](#), our supply chain survey suggests TSMC will likely expand its CoWoS capacity to 1,100kpcs in 2026F (or c.130kwpm by the end of 2026F) and bring this up to 2,000kpcs in 2027F. Although TSMC has turned more aggressive in its CoWoS plan, our contrarian view is that "WoS" (not controlled by TSMC) and many small components (e.g. IC substrates) would very likely become a bigger bottleneck than "CoW" (controlled by TSMC) in 2027F. We only model 1,800kpcs of CoWoS output in 2027F (despite our assumption of a TSMC target of 2,000kpcs).

While we have no clear bottom-up estimates about how TSMC is going to expand its CoWoS capacity beyond 2027F, we had tried to triangulate a possible trajectory in [our Asia AI Semi & Sever Anchor Report](#), based on TSMC's AI semi growth guidance and our assumptions of manufacturing content added. TSMC's guidance might hint to form 2,500-3,500kpcs of annual CoWoS capacity by 2029F, vs 680kpcs in 2025, and this would suggest a 40-50% capacity CAGR over 2025-29F compared to a >80% CAGR

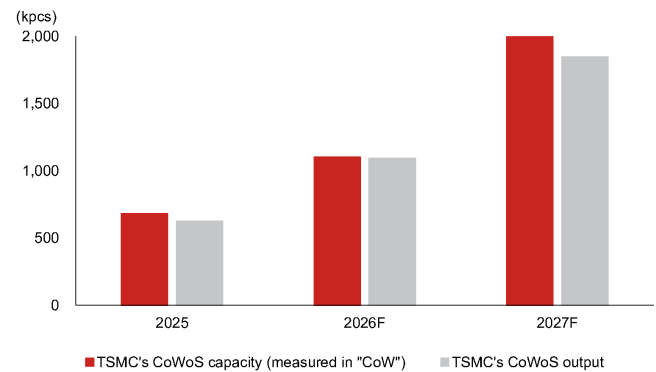
planned for 2022-27E. If the expansion track prevails, we believe Chroma's AI SLT handler sales could continue to benefit from associated backend testing capex investments by TSMC and its ecosystem partners beyond 2027F.

Fig. 24: TSMC turning more aggressive on CoW capacity expansion



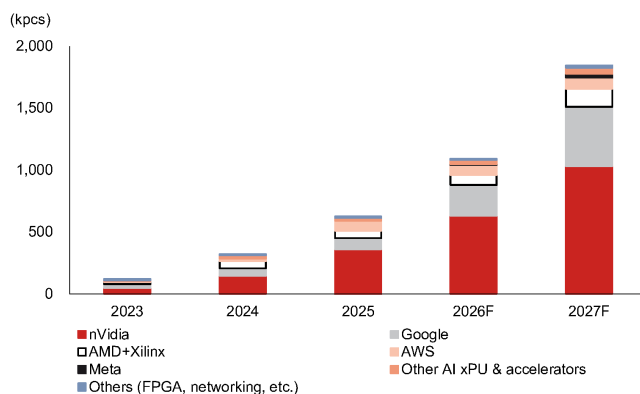
Source: Company data, Nomura estimates

Fig. 25: But the output will be constrained by "WoS"



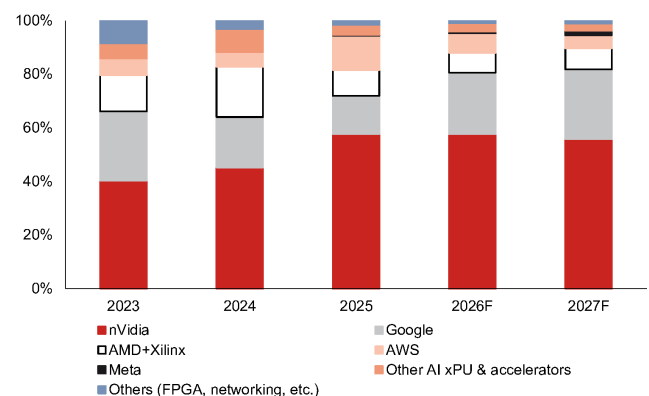
Source: Company data, Nomura estimates

Fig. 26: TSMC's CoWoS output breakdown



Source: Company data, Nomura estimates

Fig. 27: TSMC's CoWoS output allocation



Source: Company data, Nomura estimates

Emerging OSATs' 2.5D engagements bode well for Chroma in the long run

We believe that more 2.5D/CoW engagements and capacity additions by OSATs could be long-term volume catalysts for Chroma's SLT handler business, aside from TSMC's expansion, if SLT gradually becomes an indispensable norm for compute chips. We first wrote about TSMC's prudent approach to CoW capacity expansion in our *Asia AI Semi & Server Anchor report in August 2025*, and noted that such planning was critical for OSATs as it had driven most AI chip customers to look for alternative CoW suppliers. We estimate ASE (3711 TT, Buy) could form 25kwpm of FOCoS capacity by end-2026F, from 5kwpm installed by end-2025.

Amkor (AMKR US, Not rated) is also an alternate CoW partner, and management highlighted over a dozen 2.5D engagements (silicon interposer-based, as per Amkor's definition) and expected high-density fan-out RDL devices (i.e., organic interposer-based) ramping up production in 2026E and bridge-type solution for AMD in 2027E (see *Amkor's Investor Day 2026*). Our industry checks suggest Amkor might have c.15kwpm of 2.5D/SWIFT capacity by end-2026F.

Despite many 2.5D/molded interposer-based packages in the pipeline of OSATs are for CPUs owing to more relaxed technological requirements (e.g., RDL line/space) and the absence of expensive HBM content (*report*), those chip packages are naturally more complicated than conventional FCBGA and could prolong the testing time.

Fig. 28: 2.5D advanced packaging solution comparison

2.5D chip-last	TSMC	Intel Foundry	Samsung Foundry	ASE	SPIL	Amkor	Powertech
Silicon/TSV interposer	CoWoS-S (~3.3x ret.)	Foveros-S (~4x ret.)	I-CubeS H-Cube	2.5D	2.5D	2.5D	2.5D
Fan-out RDL	CoWoS-R (~5.5x ret.)	Foveros-R (production in 2027E)	n.a.	FOCoS	FO-MCM	S-SWIFT	CLIP (PLP)
Fan-out bridge (embedded in RDL)	CoWoS-L (>14x ret. by 2029E)	Foveros-B (production in 2027E)	I-CubeE	FOCoS-B	FO-EB	S-Connect	PiFO (PLP) (~9x ret. by 2028E)
Fan-out bridge (embedded in IC substrate)	-	EMIB (>12x ret. by 2028E)	-	-	-	-	-

Source: Company data, Nomura research

Fig. 29: Major CoW projects at OSATs

	Fan-out RDL	Fan-out bridge
ASE/SPIL	AMD Medusa?	AMD Venice
Amkor	nVidia GB10 nVidia Vera Microsoft Cobalt 200	AMD Venice? (2027E)
Powertech	AMD Medusa? (PLP)	AMD's next gen? (PLP)

Source: Company data, Nomura research

Google Axion CPU – a new SLT opportunity for Chroma

One example of potentially more SLT adoption is Google's own Axion CPU lineup, and a new demand driver for Chroma's equipment regime. We wrote in our [December 2025 Asia AI Semi & Server report](#) that we observed that an increasing number of CSPs were developing their own ASIC CPUs for AI assistance purposes, in addition to the typical x86 CPU general servers. We believe the trend is not at recess currently (and likely will not). The head node CPU amount paired with accelerators has been widely discussed after Google released its latest TPU v8t/v8i in late April 2026. The company specially mentioned that it has doubled the CPU usage per server, and it has increased adoption of in-house Axion CPUs – previously this was usually paired with x86 CPUs.

We had flagged demand upside from Google's ARM-based CPU "Axion N4A" (on TSMC 3nm; codenamed "Cypress") to the Asia supply chain in [August](#) and [December](#) last year, which in part is driven by the CPU adoption shift in the latest TPU 8t/8i from x86-based to Google Axion. We believe the supply chain logistics of Axion N4A are handled by GUC (3443 TT, Neutral), and KYEC (2449 TT, Buy) supports the FT stage using Advantest's ATE paired with Hon's ATC/handler, and **the SLT stage using Chroma's handler**. The current project in production has a monolithic layout, but our industry checks suggest the next generation may be a dual-die configuration, still utilizing FCBGA package, for releases in 2028F.

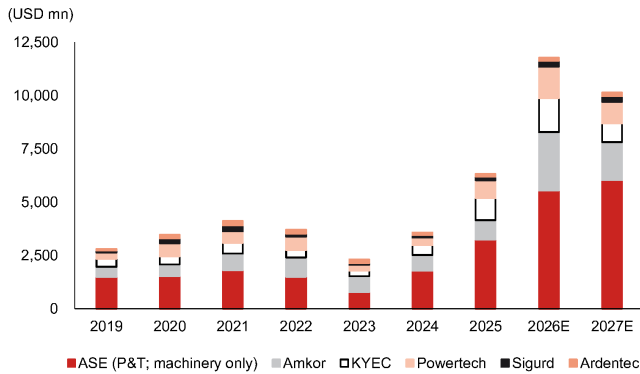
We argue that chiplet-based or multi-die floor plans will trade monolithic simplicity for a highly complex testing paradigm to validate die-to-die interconnect efficiency at longer testing time. Moreover, the multi-die layout naturally distributes the thermal load across the package and hence poses an intricate thermal management challenge to the SLT handlers. The system must deploy ultra-precise, multi-zone active thermal heads to dynamically balance the wildly mismatched self-heating conditions across dies to prevent inflicting thermal damages on the chip package. Chroma, in our view, possesses unique ATC capabilities to stand out as early as the engineering phases.

Fig. 30: An overview of CPU layout

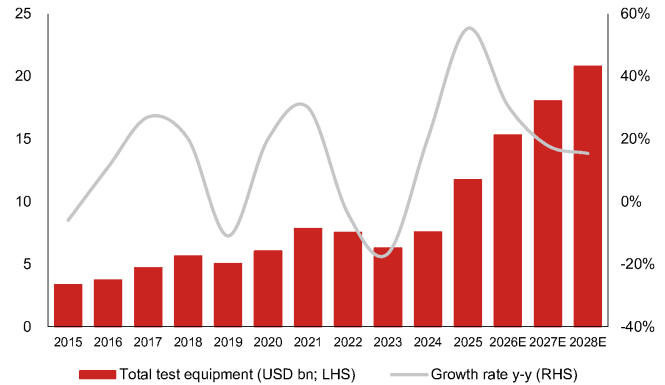
Source: Company data, Nomura estimates

Chroma to benefit from increasing testing spending by AI OSATs

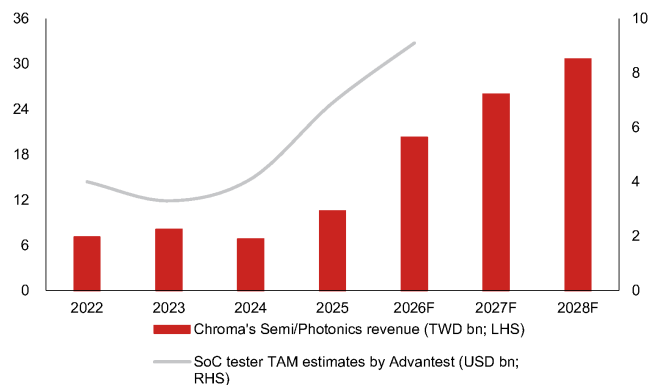
We expect Chroma's ATC/handler business to benefit from an increase in testing capex by AI OSATs as large package footprints remain the pillar of AI accelerators, and OSATs are aggressively adding floor space and tester capacity in response to the considerable surge in total test time per devices. We compile capex plans and estimates by major OSATs with testing service exposures to North American AI chip customers, which are poised to jump in 2026-27F. Although some of them do not break down spending budgets in terms of assembly and test; directionally, we expect higher dollar spent on testing, with potential upside contemplating TSMC's more aggressive CoWoS expansion plans and OSATs' participation in the full 2.5D turnkey. Meanwhile, SEMI also projects global test equipment spending to record a c.20% CAGR over 2025-28E. We believe all these should point to a robust SLT handler growth trend for Chroma through 2028F. For near-term trajectory, we compare Chroma's semi/photonics test revenue patterns to leading-edge SoC tester TAM projections provided by Advantest and Teradyne as well, despite no perfect numerical relationship between tester sales and SLT handler sales, which both suggest a robust y-y business growth in 2026F.

Fig. 31: AI OSAT capex trend

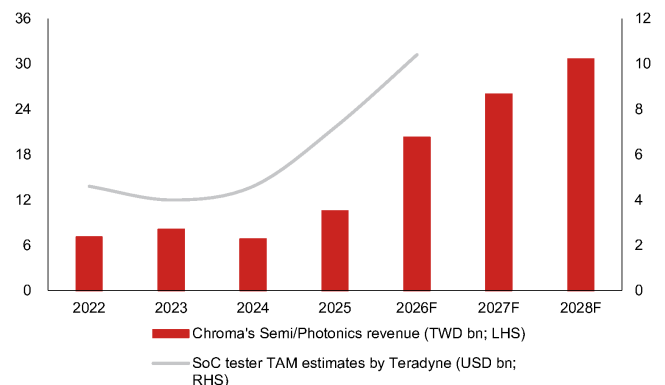
Source: Company data, Bloomberg Finance LP, Nomura estimates

Fig. 32: Increasing test equipment spending

Source: SEMI, Nomura research

Fig. 33: Chroma's revenue vs. SoC tester TAM estimates by Advantest

Source: Company data, Nomura estimates

Fig. 34: Chroma's revenue vs. SoC tester TAM estimates by Teradyne

Note: Teradyne has not yet broken down its ATE TAM of USD12-14bn in 2026E. We assume 80% of the TAM comes from SoC testers based on historical patterns.
Source: Company data, Nomura estimates

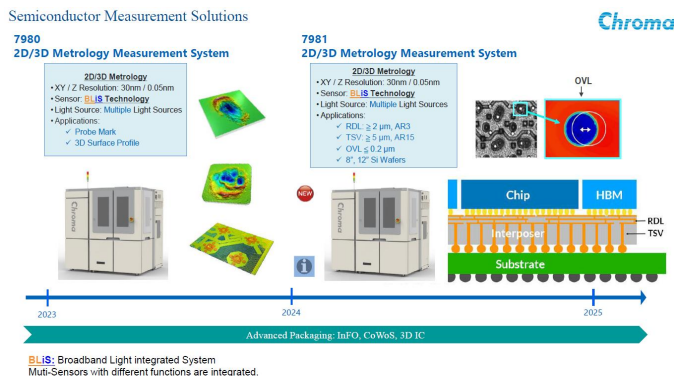
Moving up the manufacturing chain with metrology tools

Although Chroma's semiconductor test solutions have historically focused on the back end of the manufacturing flows (i.e., FT and SLT), the company has attempted to move up the value chain and tap into wafer-level metrology with Model 7980 and Model 7981, feasting on growing advanced packaging demand. We note that Model 7980 is used to inspect the microscopic impressions left on a wafer's bond pads by electrical test probes after CP, and Model 7981 runs measurement and inspection of redistribution layer (RDL) key dimensions (layer thickness, in our view), which becomes more challenging due to process variation, multi-layer build-up and finer metal interconnects. Both systems utilize Chroma's proprietary broadband light interferometry system (BLiS) to achieve 2D/3D nanoscale nondestructive optical measurement.

We believe both systems will tag along with TSMC's and OSATs' 2.5D/CoWoS capacity expansion. For Model 7980 specifically, we reckon a major demand driver could be TSMC's outsourcing of CP to OSAT partners such as KYEC which have to build up corresponding CP capacity (and therefore probe mark metrology lines). Chiplet-based designs also compound CP testing intensity for subsequent known good die assembly, and more CP cycles might further ignite demand for Chroma's Model 7980, in our view. As far as Model 7981, we view the trend toward more layers of RDL build-up in AI/HPC chips for stronger performance an incremental positive (AI/HPC's layer count is typically more than five), and the adoption of chip-last WMCM in mobile application processors may relax natural layer constraints for future upgrades (InFO-PoP has 3L of RDL, and the first-generation WMCM also features 3L). Additionally, Chroma is working on the

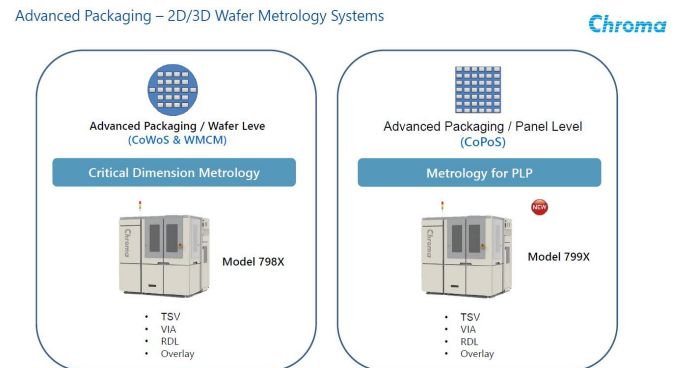
development of new wafer-level metrology tools to prepare for the roll-out of CoPoS, whose form factor change will result in another round of equipment upgrades, in our view, due to different inspection aspects for panel carrier vs. round carrier.

Fig. 35: Chroma's metrology tools



Source: Company data, Nomura research

Fig. 36: Chroma's metrology



Source: Company data, Nomura research

New opportunity from CPO test insertions

We expect photonic testing in CPO to be an additional long-term growth catalyst for Chroma's Semiconductor/Photonic Test business, and Chroma's test handlers will likely play a role in **test insertion 3** for validating optical engines (OEs) after die singulation and receptacle attach, which is a sheer optical test (O/O test). We view Chroma's established experiences in photonics, through LED, VCSEL (e.g., 3D sensing), and CMOS image sensor reliability testing, as a structural advantage paving the way for test insertion 3, where optical properties become the most critical yield determinants. Interestingly, from our industry surveys, there are two variants of test insertion 3 under development, implying **no uniform standard** for the flow at the moment. One of the approaches test the OE directly, whereas the other is a post-assembly O/O test after a fiber array unit (FAU) or a receptacle is attached onto the OE.

Chroma's experience in photonics could also grant Chroma a ticket to participate in **test insertion 4** (validation of a CPO module housing switch ASIC and OEs). In this phase, we believe Chroma will be shipping an **O/O tester for external laser source reliability** before SLT. Meanwhile, Chroma is also developing wafer-level tester solution for photonic IC wafers (i.e. insertion 1).

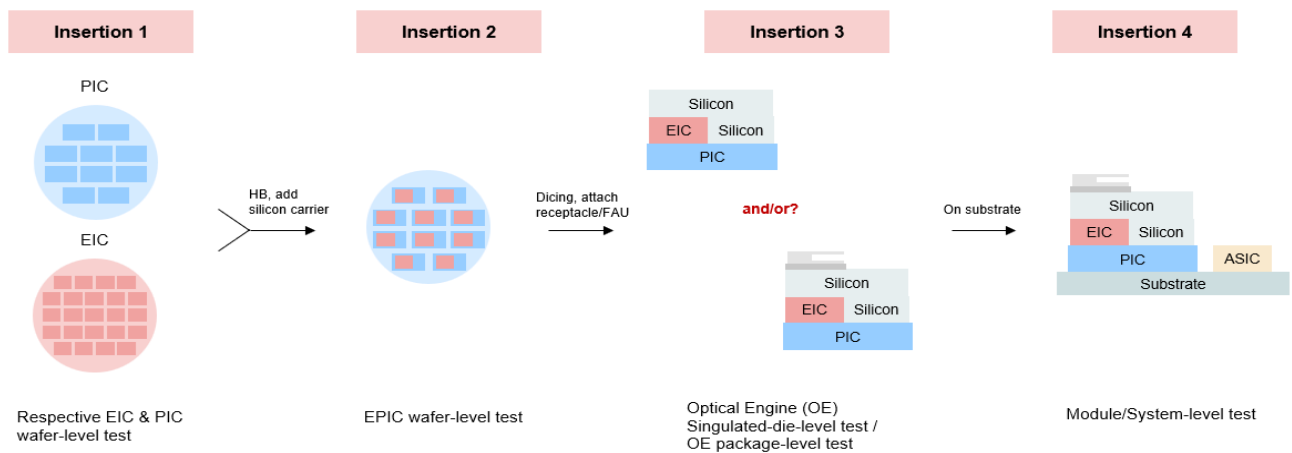
Any undetected defect in a single optical component would force manufacturers to scrap an entire expensive CPO switch module, and therefore testing cannot wait until final assembly. Strategically inserting tests early to validate known good optical/electrical dies and inspecting packages is the only way to manage the intense thermal-mechanical vulnerability of CPO and ensure viable manufacturing yield. As far as we know,, there are currently four test insertions during CPO production by TSMC and its ecosystem partners, with two occurring under the scrutiny of TSMC and the subsequent two occurring at the OSATs. We take a deep dive into test insertions in our [Anchor Report](#).

Our current expectation is Chroma will play a role in CPO test insertion 3 and insertion 4, but our model assumption only considers Chroma to begin small volume tool shipments in 2027F. Ramp-up timeline of CPO remains an uncertainty to us, and we will closely monitor the fluid supply chain dynamics. Any faster progress in the technology deployment could present upsides to Chroma's earnings beyond 2028F, in our view.

Fig. 37: CPO insertion flow and potential suppliers

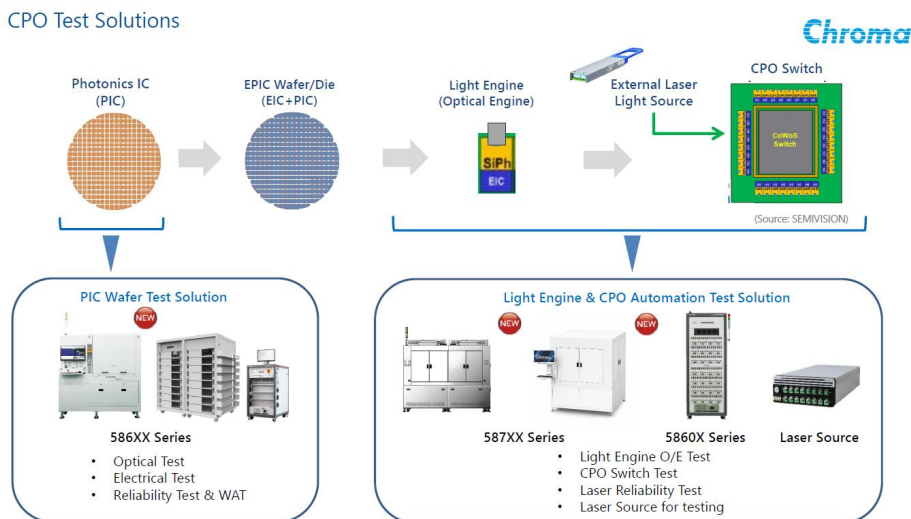
	Insertion 1	Insertion 2	Insertion 3	Insertion 4
Venue	Foundry	Foundry	OSAT	OSAT
Process	Respective EIC & PIC wafer-level test (single-sided)	EIC die on PIC wafer (EPIC) wafer-level test (double-sided)	Singulated-die-level test OE package-level test	Module/System-level test (ASIC + OE)
ATE/Instruments	Advantest/Teradyne/Ficontec/Chroma? Keysight/Viavi	Teradyne/Advantest?	Advantest/Teradyne?	Advantest?/Teradyne Keysight
Prober	FormFactor/TEL	Ficontec/MPI/FormFactor/TEL?	MPI/TEL?	
Probe card	FormFactor?	MPI/FormFactor?	MPI?	
FT/SLT Handler			Chroma/Ficontec?	Hon Chroma (ELS) WinWay
Socket			WinWay?	WinWay

Source: Nomura research

Fig. 38: CPO insertion flow

Note: This is a conceptual illustration, not specific to any company or project.

Source: Nomura research

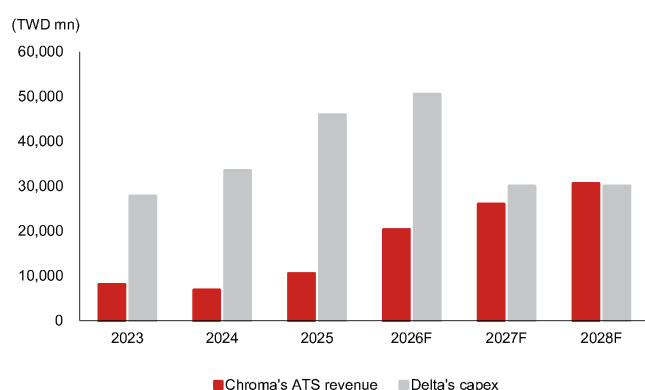
Fig. 39: Chroma's solutions in CPO test insertions

Source: Company data, Nomura research

ATS – fully charged in AI server power upgrades

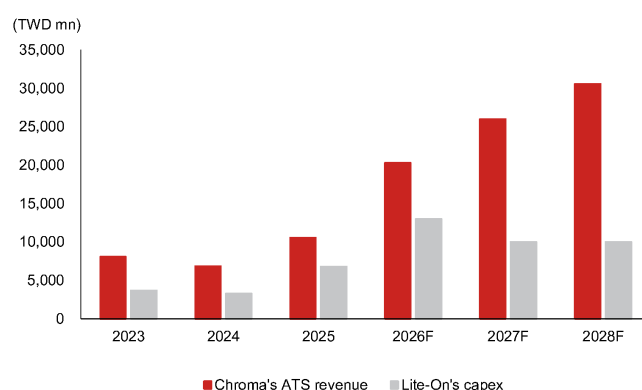
We believe Chroma is well positioned to monetize the transition of the AI server power architecture moving from traditional ~50V DC intermediate bus voltage (IBV) toward **HVDC (±400V or 800V) power racks** in view of better energy efficiency. Chroma has 38% of its revenue from ATS in 2026F and provides test and measurement equipment to power supply makers such as Delta (2308 TT, Buy) and Lite-On (2301 TT, Buy) in order to build and qualify their next-generation “powerhouses”. Chroma has comprehensive test solutions for server power supply units (PSUs), power shelves, power racks, and backup battery units (BBUs), and its product roadmap has moved in lockstep with the industry’s HVDC development journey. We expect a multi-year, capex-linked revenue stream for Chroma rather than a one-off product upgrade cycle, and therefore estimate Chroma to grow its ATS revenue at a 43% CAGR over 2025-28F against our modeled capex for Delta and Lite-On where we see potential spending upside.

Fig. 40: Chroma's ATS revenue vs. Delta's capex



Source: Company data, Nomura estimates

Fig. 41: Chroma's ATS revenue vs. Lite-On's capex



Source: Company data, Nomura estimates

AI power upgrade to HVDC is underway

Against the backdrop of rising power consumption and performance demand from AI chips, we believe the current mainstream power supply solution – 33kW and 72kW power shelves housed within IT racks and operating at 48-54V (~50V) DC as the IBV – is approaching its practical ceiling for further scale-up, constrained by rack space and rising energy losses. We believe AI racks running at TDP of 400kW and above will need to move toward **standalone power racks**, while racks operating at 750kW-1MW and above will require **800V HVDC power racks** to sustain acceptable energy efficiency, given that 800VDC offers significantly lower current and lower energy vs 50VDC. We expect accelerating AI chip performance upgrades to drive HVDC power rack adoption by late 2026F-2027F, with some CSPs potentially moving even earlier to capture the energy-efficiency and total cost of ownership (TCO) benefits. Please see [our Asia AI Power Supply report in September 2025](#) for more comprehensive color on HVDC power rack upgrades.

For nVidia's AI GPU products, GB200 NVL36 and GB300 NVL72 racks, respectively consume 130-150kW and 180-200kW per rack, and each is predominantly powered by 4 and 6 layers of 33kW 1U power shelves (based on 54V PSU, each supplying 5.5kW) with the rack. Some GB300 NVL72 configurations instead use 3-4 layers of 72kW 1U power shelves (based on 54V PSU, each supplying 12kW).

The baseline model of upcoming VR200 NVL72 rack, in our view, may draw 220kW per rack. Without a standalone power sidecar, we think the rack could be powered by 2-4 layers of 110kW 3U power shelves factoring in power redundancy, and the power shelf is built on 54V PSU, each supplying 18.3kW. The other available option is equipping the rack with 4-5 layers of 72kW 1U power shelves (based on 54V PSU, each supplying 12kW).

Despite a compromised chip floor plan for Rubin Ultra (i.e., from four GPU dies in a

package to two dies, similar to Rubin) and a likely production bottleneck with the Kyber architecture of higher compute density than the current Oberon, the TDP growth trajectory of the nVidia AI platform remains directionally intact, in our view. We may not witness a massive TDP leap in Rubin Ultra, but **the Feynman platform** (scheduled for 2028E) could be a node that sharply inflects up, considering the aggressive GPU-on-GPU SoIC stacking proposed on the chip end together with possible concurrent upgrades in other power-hungry interconnecting components. When the platform roadmap gradually unfolds even closer to 1MW per rack, only 800V HVDC power rack could adequately support the power requirement.

Delta is en route to ± 400 VDC shipments from 2H26F for a top US CSP

We are aware of recent concerns in the investment community about slower-than-expected HVDC adoption and a setback for the power supply architecture transition, with some only recognizing 800VDC as “the real HVDC”. Nevertheless, we argue the perception is biased and have been holding the view that **± 400 VDC adoption will kick off first, whereas the 800VDC timeline is subject to safety certifications**, and that ± 400 VDC is not a detour from 800VDC. We believe the adoption trajectory remains intact, and a top US CSP is en route to ± 400 VDC adoption from 2H26F with power sidecars supplied by Delta, likely beginning with VR200 racks before extending the architecture to its own ASIC racks. We expect Chroma to be a key test instrument supplier to Delta and is hence poised to benefit from ± 400 VDC moving from qualification toward volume production.

We would also note that the ± 400 VDC racks require **BBU and capacitor bank units (CBUs)**, both adding complexity to the rack architecture and the associated test requirements. The addition of BBU and CBU, in our view, suggest an extension of test coverage beyond the core power rack conversion path to include also backup power and transient energy buffering validation, thereby layering incremental test content and complexity onto an already-expanding HVDC test scope. This could present a tailwind to Chroma, with product breadth spanning across power, battery and rack-level instrumentation.

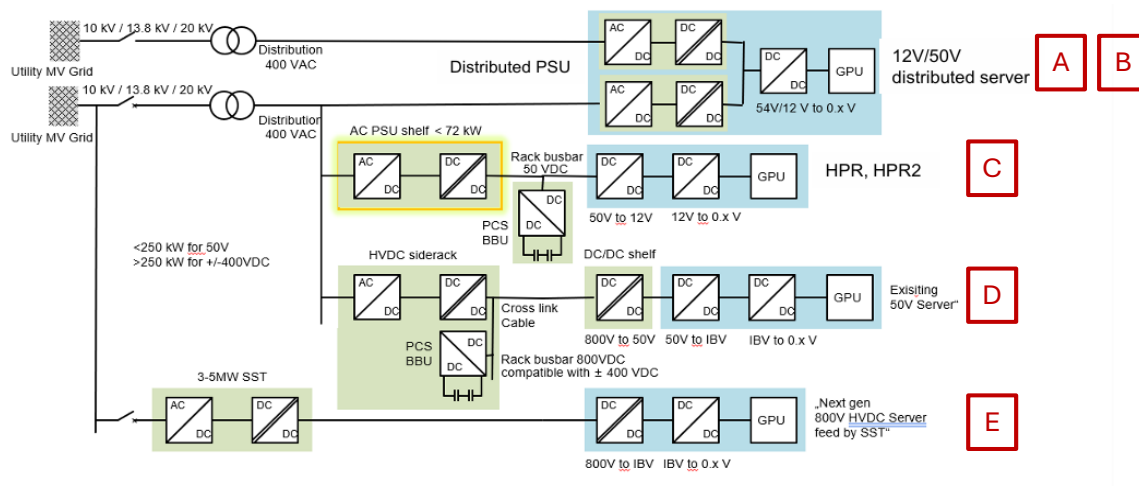
Even without HVDC, more power shelves also complicate testing efforts

In the worst case, if there are no signs of HVDC adoption in the upcoming VR200, the rack would still require more trays of PSUs to support the higher power level, and the testing complexity would therefore rise in tandem with shelf counts, in our view. Based on the aforementioned discussions about specs, the VR200 NVL72 would need a total of 6-12 x 18.3kW/4-5x 72kW PSU trays vs 6x 5.5kW/3-4x 72kW PSU trays in the GB300 NVL72. We reason that every additional power shelf layer introduces more parallel current-sharing paths and more interconnects that must be validated both independently and in aggregate under real load conditions. As a result, testing equipment has to handle more current, and engineers must verify that multiple PSUs can share load and stay in sync. More sophisticated testing efforts, in our view, remain a secular positive for Chroma.

The HVDC power evolution does not simply stop at 800V, with SST emerging

We view the **solid state transformer (SST)** as the next-generation architecture for HVDC power delivery in AI data centers. SST directly converts medium-voltage AC (MVAC, 10kV to 34.5kV utility-grade electricity) into HVDC distribution voltages (± 400 V or 800V) and integrates the traditional MV transformer and HVDC conversion functions into one unified stop. By consolidating what has historically been a multi-stage conversion process into fewer steps, SST significantly reduces cumulative power conversion losses, since the aggregate efficiency loss along the grid-to-chip power path decreases when the conversion stages decrease. We have an illustration of the SST-based pathway as Path E in *Fig. 42*.

Fig. 42: Overview of power architecture in data center



Source: Delta, Nomura Research

Beyond efficiency, we see SST as a structural enabler of the broader shift toward carbon-neutral data center design. A key architectural advantage is that SST power conversion is **bidirectional**. Traditional passive transformers rely on electromagnetic induction between primary and secondary coils (or windings) around a shared magnetic core to step voltage up or down with no active control over power flow direction (i.e., unidirectional). By contrast, an SST is built from a cascade of **active** power electronic stages using semiconductor switches (e.g., SiC or GaN-based for high-frequency switching) rather than passive magnetic coupling. The circuit topology typically starts with AC/DC rectification, followed by a high-frequency isolated DC/DC that steps voltage down to HVDC distribution levels, and with **an optional DC/AC stage for grid-interactive applications**. Because power flow direction is controllable at each active stage, the overall SST can support energy flow both from grid to storage (charging up BBUs during normal operation), and from storage back to load or grid (discharging during peak demand or grid instability) and offer an essential capability for micro-grid functionality rather than simple backup power.

Paired with energy storage systems (ESS) primarily consisting of BBUs, the bidirectional SST architecture can facilitate micro-grid integration, allowing data centers to buffer, store, and dispatch power more flexibly while smoothing grid demand and improving overall resilience. This SST+ESS combination could become a critical enabler for future-generation AI data centers, particularly as server rack loads scale to 1MW or higher, where consolidating grid-to-chip efficiency and scalability becomes essential to sustain both performance and power economies at scale.

More complicated power electronic testing moving to HVDC and SST

We reckon that AI server's shift toward 800V HVDC could reshape the entire test and measurement stack, as the current DC test regime is on the component level with unidirectional power supply, and electronic loads, power analyzers, and safety standards are all optimized for low-voltage (~50V), high-current profiles. However, when at 800V, we think the test instrumentation has to undergo structural reform because a substantially higher voltage changes **insulation, creepage/clearance, and safety requirements** in every test interface. Additionally, **bidirectional power flows** between grid, storage and load (notably SST) may need test equipment that can perform the role of source and sink vs. conventional one-way power supply. Much higher power drain by the rack could also spur the move from bench instruments to rack-scale test infrastructure.

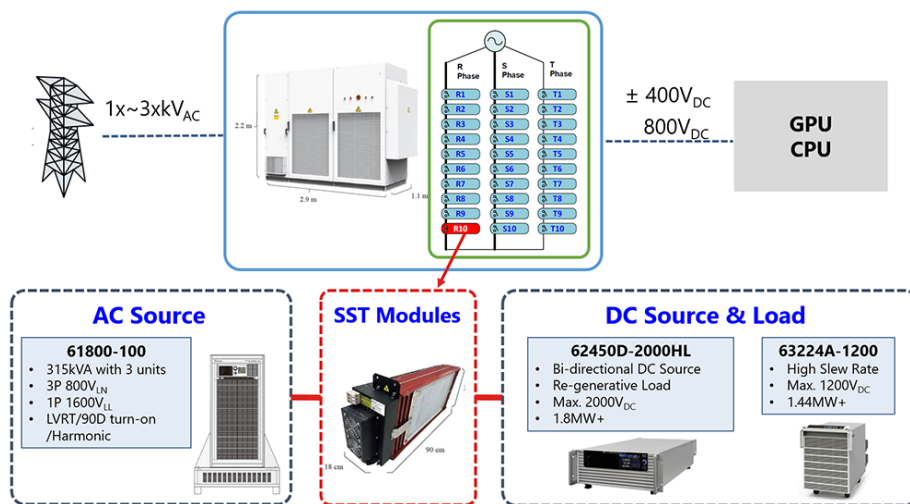
We note that a few specific parameters become more crucial under the HVDC domain:

- **Slew rate:** Slew rate is how fast a voltage or current can change, and has become

more important in HVDC test become AI/HPC workloads create extremely fast, large-magnitude load steps (e.g. kilowatts within microseconds). The test system must be capable of ramping up the loads quickly to detect and validate the HVDC bus for transient response, droop, and recovery without triggering instability.

- **Regenerative testing:** Instead of dissipating test energy as heat through resistive loads, regenerative electronic loads sink power and feed it back to the grid. Regenerative testing is becoming more standard at multi-hundred-kW test power levels, both for facility thermal load management and for energy cost savings during high-volume production testing.
- **BBU testing:** BBUs will play a strategically more important role in HVDC racks, which support more power-hungry AI racks that cannot bear the loss from transient load swings or grid interruptions. BBUs can provide short-term localized power backup during power interruptions, and current BBUs are mainly operated at 12V and 54V for data center systems. For HVDC, BBU needs to support 800V, usually through series connection of batteries. Individual cell requires its own voltage-monitoring and balancing circuit, and thus more cells suggest more validation efforts. Moreover, BBUs must support rapid bidirectional charge/discharge cycles in sync with rack-level power demand, adding transient response and synchronization testing vectors on top of standard capacity/cycle-time validation.
- **Arc testing:** An arc is an unwanted electrical discharge through the air when a circuit connection loosens under load, and lightning bolt is an example of arc. DC arcs are particularly hazardous since the current never crosses zero (according to the physical property) and DC arcs tend not to self-extinguish upon ignition, unlike AC. The arc risk increases drastically at 800V HVDC, and thus makes arc detection and fast fault interruption a critical safety protocol.
- **Double pulse test for wide-bandgap devices:** The migration to HVDC could accelerate SiC and GaN adoption in the conversion stages for better efficiency and faster switching at higher voltages. This, in turn, raises the bar for test equipment bandwidth to accurately measure switching loss and electromagnetic interference.

Fig. 43: SST and Chroma's power test solutions



Source: "2025 OCP APAC Summit" (CC BY-SA 4.0). From author [Peter Barbosa, DELTA](#). Rearranged and adapted by Chroma ATE Inc.

Source: Company data, Nomura research

Chroma leads in power electronics test instruments

The global electronic test and measurement market features five broader-line vendors: Keysight Technologies (KEYS US, Not rated), Rohde & Schwarz (unlisted), Tektronix (unlisted; a subsidiary of Ralliant [RAL US, Not rated]), National Instruments (unlisted; a subsidiary of Emersion [EMR US, Not rated]), and Anritsu (6754 JP, Neutral).

- Keysight is a general-purpose electronic test leader with the broadest portfolio, spanning RF/microwave, network and protocol test, semiconductor parametric test,

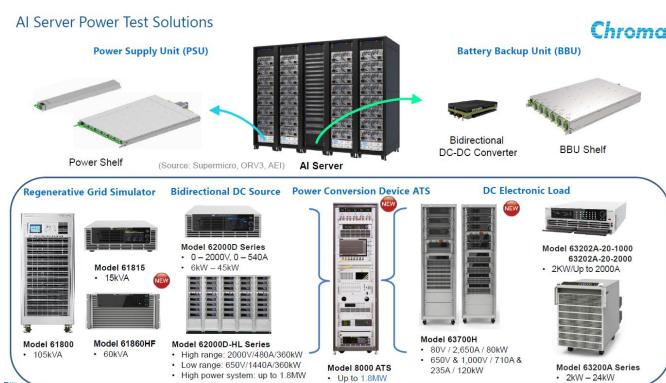
and general-purpose power analysis/electronic loads.

- Rohde & Schwarz centers around RF/wireless, electromagnetic interference and compatibility (EMI/EMC), and automotive/aerospace communications test.
- Tektronix's core is oscilloscopes, mixed-signal and power electronics analysis tools, across aerospace/defense, automotive, and renewable energy applications.
- National Instruments has a solid foothold in modular, software-defined instrumentation and highly customizable automated test systems, addressing mostly semiconductor production test.
- Anristu anchors its core to RF/microwave and optical communications test.

Chroma occupies a **distinct niche** relative to the top five vendors, and its portfolio centers around power electronics and batteries. The divergence from the global top test and measurement instrument makers per se, in our view, is strategically a competitive advantage of Chroma. The company's relevance in the AI server realm comes specifically from its high-power electronic loads, regenerative test systems, and rack/power-shelf-level tester across a wide spectrum of power levels. We notice Chroma's differentiation is less on any single specs but more **on the breadth** of its power electronics/battery test offerings coupled with **deep engineering** in high-power, bidirectional/regenerative systems, as evidenced by its continuous platform expansion including introduction of a comprehensive SST module test portfolio covering AC-side simulation and DC-side load testing (*news*).

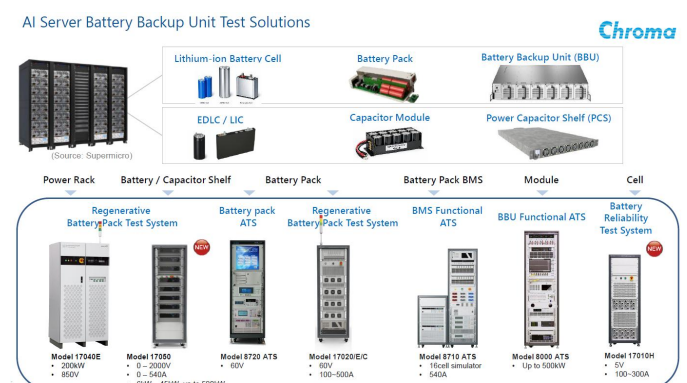
Nevertheless, we think most peers in this niche power test space compete within a narrower slice of the value chain. Kikusui (6912 JP, Not rated), ITECH (unlisted), and B&K Precision (unlisted) are positioned mainly around genera-purpose power supplies and electronic loads, while Arbin (unlisted), Maccor (unlisted), Bitrode (unlisted), and Neware (unlisted) are dedicated to battery cyclers. Chroma, by contrast, supports power supplies, electronic loads, regenerative/bidirectional systems and battery test under a single roof. As such, customers could enjoy a one-stop shop for test instruments and systems at a lower integration risk and TCO, when they have to broaden the test scope across power electronics and battery in response to structural migration in server power architectures. The end-to-end capabilities on the full power test stack with proven engineering also better help customers progress with the technical development, in our view.

Fig. 44: Chroma's AI server power test solutions



Source: Company data, Nomura research

Fig. 45: Chroma's solutions for AI server BBU test



Source: Company data, Nomura research

Earnings forecasts

We expect a 45% net earnings CAGR over 2025-28F

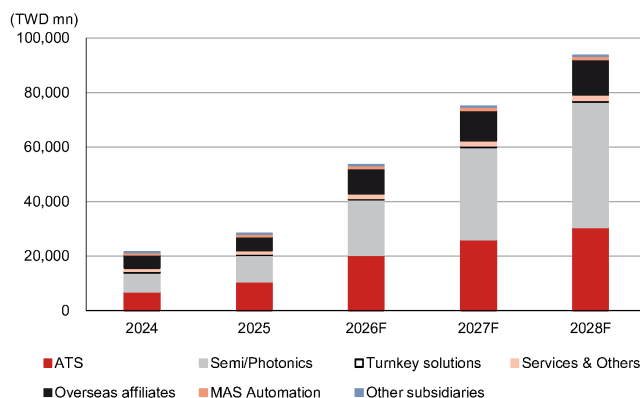
We expect Chroma to grow its revenue by 89% y-y in 2026F on the back of 110% growth in Semiconductor/Photonics and 92% in ATS, driven by continued capacity expansion for the Asia AI OSAT supply chain and tool upgrades for nVidia's AI GPU platform refresh, and initial wave of HVDC adoption starting VR200 with more stringent power test requirements. Chroma's progress into 2D/3D wafer metrology could also capture another adjacency of advanced packaging capacity growth at TSMC and OSATs, in our view. We expect Chroma's revenue to witness a 49% CAGR over 2025-28F given structural testing trend for more complexity and longer time in AI/HPC chips (notably nVidia's planned adoption of GPU-on-GPU SoIC stacking for the Feynman platform), broader adoption of SLT, and sustained tool upgrades by power supply makers to foster the industry migration toward HVDC and SST.

The initial equipment opportunities from CPO test insertions (albeit likely small) might kick in from 2027F in the earliest, but we have not yet contemplated the contribution in our earnings estimates in detail, as the supply chain dynamics remain fluid, and we have to continue monitoring the development and viability of new technologies.

On the profitability front, we expect rather stable GM profiles across Chroma's major product lines, and believe equipment upgrades driven by power electronics and semiconductor testing are the predominant factors to the company's GM upswings. Our current assumptions reflect continued upgrade cycles along with AI/HPC and power electronics customers' roadmap primarily. We believe CPO test insertion tools could also be accretive to Chroma's GM in the longer run. We, hence, model GM of 62.4%/62.9%/63.7% in 2026-28F vs. 61.5% in 2025.

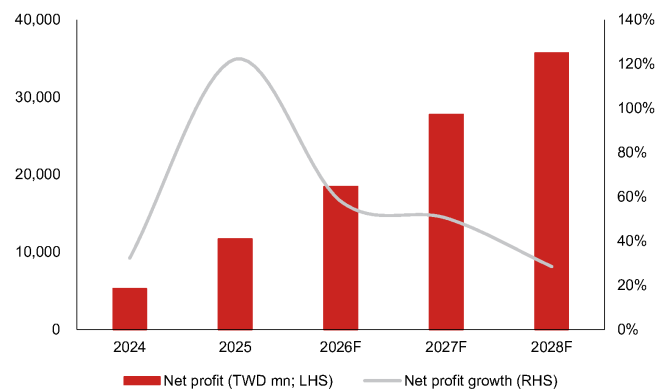
Blended in a 25% opex CAGR over 2025-28F, we anticipate Chroma to enjoy operating leverage from rapid topline expansion and grow its OPM to 46.3% in 2028F from 32.5% in 2025. Net, we estimate the company to deliver a 45% EPS CAGR over 2025-28F and think a successful ramp-up of CPO test insertion beyond 2028F could drive a more significant earnings upgrade.

Fig. 46: Chroma's revenue breakdown



Source: Company data, Nomura estimates

Fig. 47: Chroma's net profit and y-y growth



Source: Company data, Nomura estimates

Fig. 48: Chroma's P&L

(TWD mn)	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2025	2026F	2027F	2028F
Revenue	6,865	6,455	6,410	8,580	11,860	13,529	14,157	13,968	15,539	18,295	20,325	20,820	28,311	53,514	74,977	93,623
Gross profit	4,148	4,224	3,835	5,217	7,424	8,442	8,834	8,716	9,773	11,509	12,788	13,100	17,425	33,416	47,171	59,617
- Opex	(1,984)	(2,027)	(1,987)	(2,229)	(2,627)	(2,706)	(2,831)	(2,794)	(2,875)	(3,293)	(3,618)	(3,706)	(8,227)	(10,957)	(13,491)	(16,232)
Operating profit	2,164	2,197	1,848	2,988	4,797	5,736	6,003	5,922	6,899	8,216	9,170	9,394	9,198	22,458	33,680	43,385
Pretax profit	2,597	2,438	5,475	3,408	5,094	5,919	6,231	6,178	7,162	8,491	9,454	9,659	13,918	23,421	34,767	44,623
Net profit	2,122	1,953	5,066	2,550	3,864	4,708	4,966	4,932	5,719	6,784	7,555	7,720	11,692	18,471	27,778	35,687
EPS (TWD)	5.03	4.63	11.99	6.04	9.12	11.11	11.72	11.64	13.50	16.01	17.83	18.22	27.70	43.59	65.56	84.23
Profitability	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2025	2026F	2027F	2028F
Gross margin	60.4%	65.4%	59.8%	60.8%	62.6%	62.4%	62.4%	62.4%	62.9%	62.9%	62.9%	62.9%	61.5%	62.4%	62.9%	63.7%
- Opex ratio	-28.9%	-31.4%	-31.0%	-26.0%	-22.1%	-20.0%	-20.0%	-20.0%	-18.5%	-18.0%	-17.8%	-17.8%	-29.1%	-20.5%	-18.0%	-17.3%
Operating margin	31.5%	34.0%	28.8%	34.8%	40.5%	42.4%	42.4%	42.4%	44.4%	44.9%	45.1%	45.1%	32.5%	42.0%	44.9%	46.3%
Pretax margin	37.8%	37.8%	85.4%	39.7%	43.0%	43.7%	44.0%	44.2%	46.1%	46.4%	46.5%	46.4%	49.2%	43.8%	46.4%	47.7%
Net margin	30.9%	30.3%	79.0%	29.7%	32.6%	34.8%	35.1%	35.3%	36.8%	37.1%	37.2%	37.1%	41.3%	34.5%	37.0%	38.1%
Q-Q	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F				
Revenue	13.7%	-6.0%	-0.7%	33.9%	38.2%	14.1%	4.6%	-1.3%	11.2%	17.7%	11.1%	2.4%				
Gross profit	15.0%	1.8%	-9.2%	36.0%	42.3%	13.7%	4.7%	-1.3%	12.1%	17.8%	11.1%	2.4%				
Operating profit	38.8%	1.5%	-15.9%	61.7%	60.6%	19.6%	4.7%	-1.3%	16.5%	19.1%	11.6%	2.4%				
Pretax profit	33.9%	-6.1%	124.5%	-37.8%	49.5%	16.2%	5.3%	-0.8%	15.9%	18.6%	11.3%	2.2%				
Net profit	43.8%	-8.0%	159.4%	-49.7%	51.5%	21.8%	5.5%	-0.7%	16.0%	18.6%	11.4%	2.2%				
Y-Y	1Q25	2Q25	3Q25	4Q25	1Q26	2Q26F	3Q26F	4Q26F	1Q27F	2Q27F	3Q27F	4Q27F	2025	2026F	2027F	2028F
Revenue	55.4%	17.1%	13.8%	42.1%	72.8%	109.6%	120.9%	62.8%	31.0%	35.2%	43.6%	49.1%	31.0%	89.0%	40.1%	24.9%
Gross profit	61.6%	29.3%	16.0%	44.6%	79.0%	99.8%	130.3%	67.1%	31.6%	36.3%	44.8%	50.3%	36.7%	91.8%	41.2%	26.4%
Operating profit	140.3%	45.1%	22.5%	91.6%	121.7%	161.0%	224.8%	98.2%	43.8%	43.2%	52.8%	58.6%	67.8%	144.2%	50.0%	28.8%
Pretax profit	113.0%	35.1%	213.7%	75.8%	96.1%	142.7%	13.8%	81.3%	40.6%	43.5%	51.7%	56.4%	107.5%	68.3%	48.4%	28.3%
Net profit	122.2%	38.8%	255.3%	72.8%	82.1%	141.1%	-2.0%	93.4%	48.0%	44.1%	52.1%	56.5%	122.1%	58.0%	50.4%	28.5%

Source: Company data, Nomura estimates

Fig. 49: Chroma: Segment revenue forecasts

TWD mn	2024	2025	2026F	2027F	2028F
Total revenue	21,604	28,311	53,514	74,977	93,623
y-y growth	16%	31%	89%	40%	25%
Consolidated testing equipment business	20,418	27,166	52,155	73,525	92,165
y-y growth	15%	33%	92%	41%	25%
MAS Automation	873	884	1,096	1,190	1,195
y-y growth	32%	1%	24%	9%	0%
Other subsidiaries	313	261	263	263	263
y-y growth	36%	-17%	1%	0%	0%
Consolidated testing equipment business					
Parent	15,651	22,012	42,892	62,425	79,245
y-y growth	25%	41%	95%	46%	27%
Test instruments & Automatic testing system (ATS)	6,825	10,546	20,290	26,000	30,600
y-y growth	-16%	55%	92%	28%	18%
Semiconductor/Photonics test solutions	6,973	9,759	20,462	33,900	46,000
y-y growth	136%	40%	110%	66%	36%
Turnkey solutions	685	444	448	520	580
y-y growth	51%	-35%	1%	16%	12%
Services & Others	1,168	1,263	1,692	2,005	2,065
y-y growth	13%	8%	34%	18%	3%
Overseas affiliates	4,767	5,154	9,263	11,100	12,920
y-y growth	-9%	8%	80%	20%	16%

Source: Company data, Nomura estimates

Fig. 50: Nomura forecasts vs Bloomberg consensus for 2026-28F

(TWD mn)	2026F			2027F			2028F		
	NMR	BBG	Diff.	NMR	BBG	Diff.	NMR	BBG	Diff.
Revenue	53,514	52,164	2.6%	74,977	73,834	1.5%	93,623	88,504	5.8%
Gross profit	33,416	32,946	1.4%	47,171	47,127	0.1%	59,617	55,613	7.2%
Operating profit	22,458	21,443	4.7%	33,680	33,611	0.2%	43,385	45,708	-5.1%
Pretax profit	23,421	23,306	0.5%	34,767	27,751	25.3%	44,623	40,124	11.2%
Net profit	18,471	18,088	2.1%	27,778	27,975	-0.7%	35,687	32,713	9.1%
EPS (TWD)	43.59	42.32	3.0%	65.56	65.41	0.2%	84.23	77.43	8.8%
Margin	NMR	BBG	Diff.	NMR	BBG	Diff.	NMR	BBG	Diff.
Gross margin	62.4%	63.2%	-71bps	62.9%	63.8%	-91bps	63.7%	62.8%	84bps
Operating margin	42.0%	41.1%	86bps	44.9%	45.5%	-60bps	46.3%	51.6%	-530bps
Pretax margin	43.8%	44.7%	-91bps	46.4%	37.6%	878bps	47.7%	45.3%	233bps
Net margin	34.5%	34.7%	-16bps	37.0%	37.9%	-84bps	38.1%	37.0%	116bps

Source: Bloomberg Finance L.P. consensus, Nomura estimates

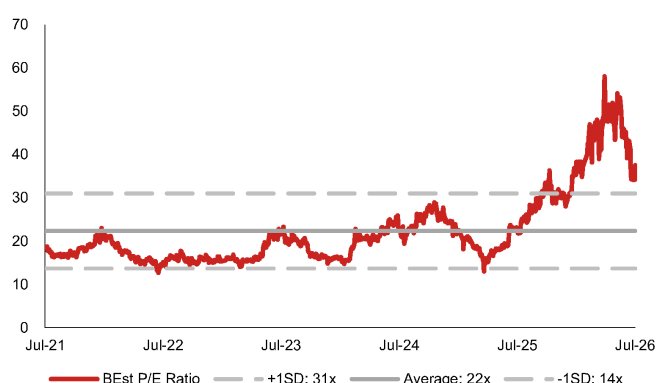
Valuation methodology and risks

We derive our target price of TWD2,845, based on 38x average 2027-28F EPS of TWD74.9. Our target P/E multiple is at the higher end of Chroma's historical trading band of 14-58x, which we view reasonable given our modeled net earnings CAGR of 39% over 2026-28F. We believe the dual growth engines from Chroma's Semiconductor/Photonic Test and ATS, propelled by a myriad of margin-accretive catalysts such as SLT handler upgrades from more complicated merchant AI GPU designs, expanding SLT footprints in more AI/HPC customers, and the transition of AI server power architecture will continue to drive upsides to earnings. Although we do not factor in contribution from CPO test insertions, a successful ramp-up of CPO and associated test insertion adoption could drive further upgrades to earnings estimates beyond 2028F.

We select Chroma's valuation comparison peers from the semiconductor backend equipment and test interface vendor space, as well as players in data center power system and power system components, given its exposures to both fields. Chroma is currently trading at 32x 2027F EPS and 25x 2028F EPS, respectively, and we believe this is undemanding compared to average valuation of 39x by semiconductor backend equipment (specifically 37x by test equipment), 35x by test interface vendor, and 27x by data center power-related players (Chroma could have a higher valuation than pure power plays due to its exposure to semiconductor equipment, in our view), all based on consensus 2027E EPS.

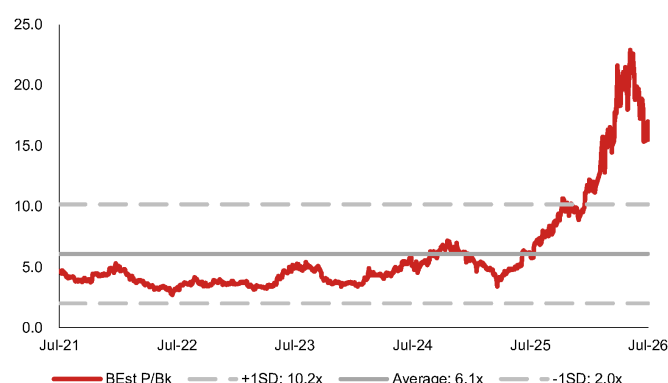
Major downside risks to our call include: 1) CoWoS and backend testing capacity expansion slowdown; 2) slower-than-expected product refresh, platform performance upgrade, or ramp-up by the AI chip end customers; 3) fiercer-than-expected from other international testing equipment makers; and 4) weaker-than-expected end-market demand, particularly for AI servers.

Fig. 51: Chroma's consensus P/E ratio



Source: Bloomberg Finance L.P., Nomura research

Fig. 52: Chroma's consensus P/B ratio



Source: Bloomberg Finance L.P., Nomura research

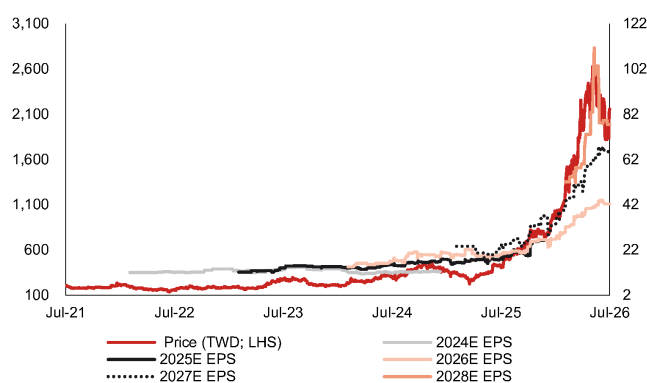
Fig. 53: Valuation comparison of Chroma

Ticker	Company Name	Rating	Last close	Target price	Upside (Downside)		Market	P/E (x)			P/B (x)			ROE			Dividend yield		
			LCY	LCY	%	Cap. USD mn	FY26F	FY27F	FY28F	FY26F	FY27F	FY28F	FY26F	FY27F	FY28F	FY26F	FY27F	FY28F	
Semi test equipment																			
6857 JP	Advantest	Buy	29,585.0	30,600.0	3	132,836	44.2	34.4	27.7	20.4	15.5	11.9	52.5%	51.2%	48.7%	0.2%	0.2%	0.2%	
TER US	Teradyne	Not rated	374.0	n.a.	n.a.	58,553	51.1	37.2	26.7	16.2	13.1	10.0	37.5%	41.7%	41.9%	0.1%	0.1%	0.2%	
KEYS US	Keysight	Not rated	328.1	n.a.	n.a.	56,072	32.2	27.6	24.6	8.2	6.8	5.6	25.9%	26.4%	26.1%	0.0%	0.0%	0.0%	
7769 TT	Hon Precision	Buy	6,405.0	11,100.0	73	35,597	47.7	29.3	19.0	16.3	12.4	9.1	37.6%	48.1%	55.3%	1.5%	2.4%	3.7%	
300604 CH	Changchuan	Not rated	309.1	n.a.	n.a.	28,944	94.5	64.3	46.8	29.3	20.5	14.5	33.4%	34.7%	33.7%	0.1%	0.2%	0.2%	
2360 TT	Chroma	Buy	2,125.0	2,845.0	34	27,912	48.7	32.4	25.2	21.3	15.7	12.3	49.8%	55.8%	54.6%	1.4%	2.2%	2.8%	
688200 CH	Accotest	Not rated	369.0	n.a.	n.a.	10,926	94.8	75.0	58.8	14.2	13.4	11.3	16.2%	17.8%	19.1%	0.3%	0.4%	0.6%	
COHU US	Cohu	Not rated	56.2	n.a.	n.a.	2,652	95.8	38.3	26.4	3.5	3.2	2.9	-207.6%	27.2%	n.a.	n.a.	n.a.	n.a.	
AEM SP	AEM	Not rated	9.0	n.a.	n.a.	2,235	42.4	27.7	21.0	5.2	4.4	3.7	13.0%	16.8%	18.9%	0.6%	0.9%	1.2%	
6337 JP	TESEC	Not rated	2,339.0	n.a.	n.a.	80	20.3	18.8	16.3	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	4.3%	4.3%	4.3%	
Average							57.2	38.5	29.3	14.9	11.7	9.0							
Semi test interface																			
TPRO IM	Technoprobe	Not rated	31.2	n.a.	n.a.	22,854	71.9	47.2	37.0	13.2	10.5	8.3	19.6%	22.6%	23.1%	0.0%	0.1%	0.2%	
6223 TT	MPI	Buy	6,000.0	8,000.0	33	18,159	100.5	46.8	26.4	31.0	20.5	13.1	35.7%	54.8%	63.1%	0.5%	1.2%	2.1%	
FORM US	FormFactor	Not rated	113.8	n.a.	n.a.	8,867	46.1	35.7	39.4	n.a.	n.a.	n.a.	15.4%	18.7%	n.a.	0.0%	0.0%	n.a.	
6515 TT	WinWay	Buy	6,500.0	8,315.0	28	7,235	73.6	35.0	19.5	32.7	20.6	12.8	46.4%	71.6%	80.4%	1.0%	2.2%	3.9%	
6510 TT	CHPT	Suspended	2,710.0	n.a.	n.a.	2,745	42.2	22.7	15.9	8.2	6.3	n.a.	20.8%	31.8%	56.0%	1.3%	2.3%	2.9%	
COHU US	Cohu	Not rated	56.2	n.a.	n.a.	2,652	95.8	38.3	26.4	3.5	3.2	2.9	-207.6%	27.2%	n.a.	n.a.	n.a.	n.a.	
6941 JP	Yamaichi	Not rated	9,380.0	n.a.	n.a.	1,181	17.6	15.7	13.7	3.2	2.8	2.4	19.3%	20.4%	20.7%	1.7%	1.8%	2.1%	
6683 TT	KSMT	Not rated	1,220.0	n.a.	n.a.	1,038	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	n.a.	
Average							64.0	34.5	25.5	15.3	10.6	7.9							
AI OSAT																			
3711 TT	ASE	Buy	656.0	730.0	11	90,585	37.2	25.5	20.2	7.1	6.2	5.5	19.7%	25.6%	28.6%	1.9%	2.8%	3.5%	
AMKR US	Amkor	Not rated	66.7	n.a.	n.a.	16,541	31.9	27.5	23.1	3.3	3.0	2.6	11.2%	11.1%	12.7%	0.5%	0.5%	0.6%	
2449 TT	KYEC	Buy	295.0	390.0	32	11,142	30.3	19.1	14.3	5.7	4.5	3.5	21.0%	26.4%	27.3%	0.3%	0.5%	0.7%	
6239 TT	Powertech	Not rated	294.0	n.a.	n.a.	6,894	24.5	16.9	15.0	3.4	3.0	2.7	15.7%	18.8%	18.6%	2.4%	3.3%	3.7%	
Average							31.0	22.3	18.1	4.9	4.2	3.6							
Datacenter power																			
SU FP	Schneider	Not rated	268.8	n.a.	n.a.	177,387	27.2	23.2	20.2	5.7	5.0	4.4	21.3%	22.2%	22.4%	1.7%	1.9%	2.1%	
ETN US	Eaton	Not rated	403.0	n.a.	n.a.	156,465	30.2	25.9	22.6	7.0	6.1	5.5	23.5%	24.5%	25.3%	1.1%	1.1%	1.2%	
2308 TT	Delta	Buy	1,880.0	2,800.0	49	150,838	44.6	27.8	20.1	14.6	10.7	8.0	36.3%	44.4%	45.5%	1.1%	1.8%	2.5%	
VRT US	Vertiv	Not rated	304.5	n.a.	n.a.	116,961	47.0	36.0	28.0	19.7	13.8	9.7	48.8%	46.4%	42.8%	0.1%	0.1%	0.2%	
FLEX US	Flextronics	Not rated	127.4	n.a.	n.a.	46,674	28.7	18.7	13.7	8.2	6.6	4.3	32.2%	43.6%	n.a.	n.a.	n.a.	n.a.	
2301 TT	Lite-On	Buy	218.0	285.0	31	15,596	22.4	15.3	12.6	5.0	4.4	n.a.	23.4%	30.5%	33.0%	3.8%	5.5%	n.a.	
002600 CH	Lead Wealth	Not rated	12.1	n.a.	n.a.	15,239	28.2	18.7	14.8	n.a.	1.2	1.1	11.5%	14.7%	16.3%	0.4%	0.7%	0.9%	
002851 CH	Megmeet	Not rated	131.7	n.a.	n.a.	11,372	101.0	48.0	33.7	10.0	8.5	7.1	10.5%	17.9%	21.4%	0.1%	0.3%	0.4%	
Average							41.2	26.7	20.7	10.0	7.0	5.7							

Note: Priced as of 22 July 2026. Bloomberg consensus for Not rated and Rating Suspended stocks.

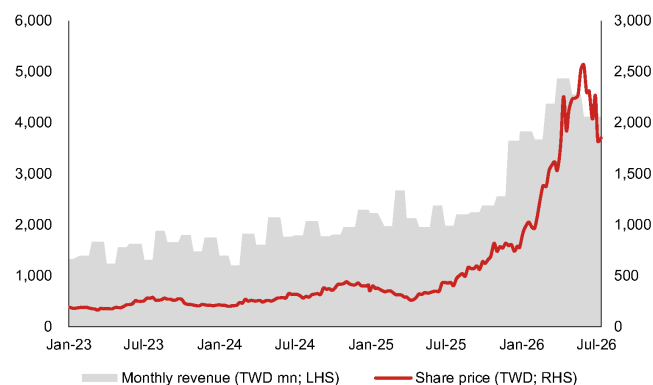
Source: Company data, Bloomberg Finance L.P., Nomura estimates

Fig. 54: Chroma's share price vs. Bloomberg consensus EPS revisions



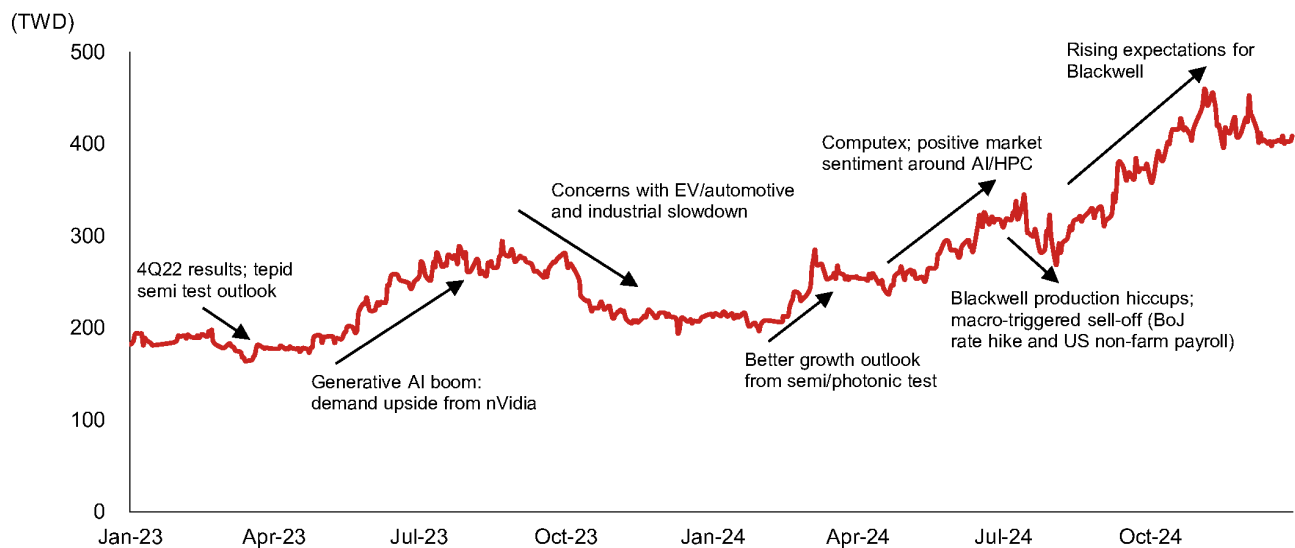
Source: Company data, Bloomberg Finance L.P. consensus estimates, Nomura research

Fig. 55: Chroma's share price vs. monthly revenue



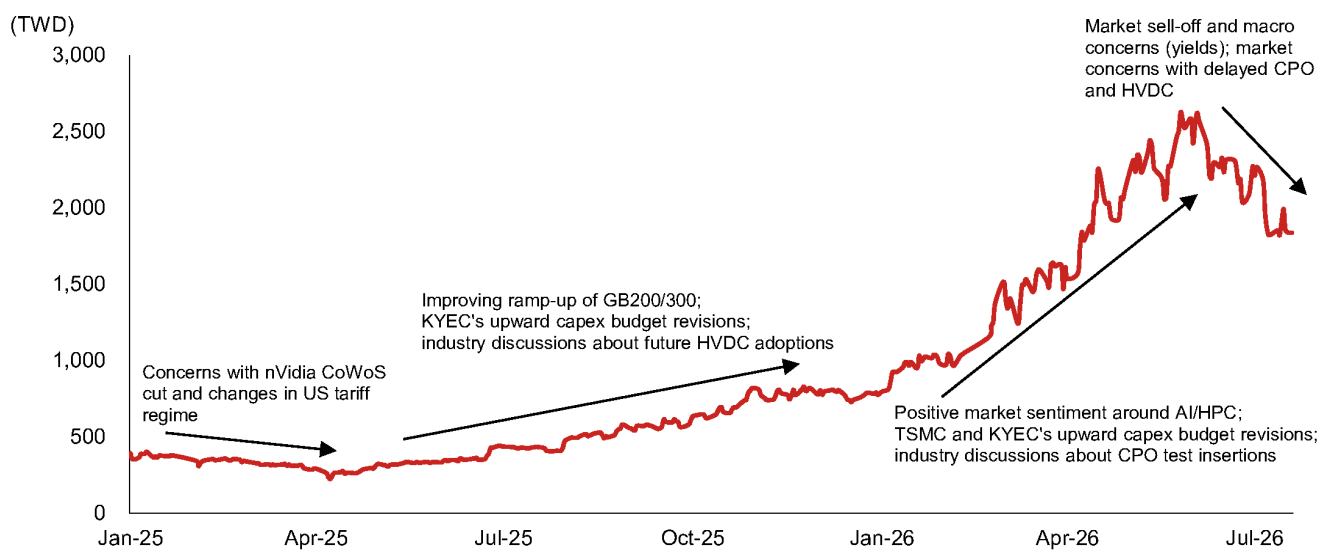
Source: Company data, TEJ, Nomura research

Fig. 56: Chroma's share price catalysts recap (2023-24)



Source: Company data, TEJ, Nomura research

Fig. 57: Chroma's share price catalyst recap (2025-26 YTD)



Source: Company data, TEJ, Nomura research

Company background

Founded in 1984, Chroma ATE is a Taiwan-based supplier of precision electronic test and measurement instruments, automated test systems, intelligent manufacturing systems, and automation turnkey solutions. Chroma's primary production bases are concentrated in Taiwan, and it also operates assembly and module production footprints in China to support localized system integration. The company's end applications span sectors where precision quality control and reliable electro-optical performance are critical, including semiconductors, EV/renewable energy (e.g., battery cells), power electronics, specialized photonics, flat panel displays (FPD), and automated optical inspection systems.

More than 95% of Chroma's revenue comes from its testing equipment business, under which are test instruments and automatic testing systems (ATS), semiconductor/photonic tests, turnkey solutions, services and others, and overseas affiliates, and ~5% from its subsidiary MAS Automation.

- **Test instrument & automatic testing systems (ATS; 37% of 2025 revenue):** This segment covers testing equipment solutions for power supplies, battery cells, battery modules, electric drivetrains (e.g., battery storage systems, DC/DC converters), EV supply equipment (EVSE), etc.
- **Semiconductor/Photonic tests (35% of 2025 revenue):** Chroma's semiconductor testing equipment solutions address FT (tester and FT handler) and SLT (SLT handler). Chroma's SoC testers address a variety of mature-node chip applications including consumer electronics (MCU, audio, PC peripherals), PMIC, RF/wireless IC, and image sensors, and its FT handlers offer chip package pick-and-place automation and thermal control capabilities to work alongside testers. In SLT handlers, Chroma focuses on niche, quality-dependent applications such as AI/HPC and automotive to help customers verify their chips against simulated real-world conditions. The other part of this revenue segment is photonic test solutions for optoelectronic components (e.g., laser diode and LED), where Chroma offers both wafer level and package level test equipment.
- **Overseas affiliates (22% of 2025 revenue):** This segment contains Chroma's revenue booked through non-Taiwan entities. We think the application base is similar to that of ATS.
- **Service and others (5% of 2025 revenue):** Chroma books recurring aftermarket service revenues and are some smaller product lines like FPD test, video and color test under this segment.
- **Turnkey solutions (3% of 2025 revenue):** Chroma provides system-integration business bundling instruments, automations and software into complete production-line solutions. Sub-segments cover photovoltaics, battery formation and testing, and EV. The nature of the line is more customized and lumpier in revenue timing.
- **MAS Automation (5% of 2025 revenue):** This subsidiary involves in automation equipment for clean-energy manufacturing, such as photovoltaic equipment and lithium battery formation and grading equipment.

Chroma's business expansion has not been purely organic. Management has run a disciplined, phased M&A or equity investment program since the company was founded to convert the company from a standalone test instrument vendor to an integrated test, automation, and turnkey solution provider with progress into upstream semiconductor testing adjacencies.

In 2019, Chroma entered into an agreement with Camtek (CAMT US, Not rated), an Israeli semiconductor metrology and inspection equipment maker, to acquire 20.5% ownership. In addition to this investment, Chroma and Camtek entered into an agreement in which Camtek would license its triangulation technology for metrology to Chroma for non-semiconductor applications, and both parties agreed to cooperate in potential semiconductor equipment projects based on synergies between inspection and metrology technologies.

We summarize Chroma's development milestones and critical M&A history in *Fig. 58*.

Fig. 58: Chroma's milestones and M&A history

Year	Milestone and M&A history
1984	- Leo Huang, Chun-Sheng Chen, Ming-Hsiung Chang, and Hsiao-Hui Huang co-founded Chroma ATE Inc. and started developing measuring instruments, including the very first product "video pattern generator (VPG)" for monitor testing.
1996	- Chroma went public in Taiwan.
1997	- Invested in DynaScan Technology, a provider of high-brightness display solutions.
1998	- Invested in ADLINK Technology, an industrial PC maker.
2000	- Invested in EVT Technology. - Acquired Zentech Technology.
2006	- Spun off Special Material Business Unit and established a subsidiary "Chroma New Material" as the agency of Nippon Micrometal to sell bonding wire and micro solder ball for semi packages.
2007	- Acquired MAS Automation, an automation equipment supplier for solar modules and Li-ion batteries. - Acquired Sajat Technology, a smart manufacturing platform provider, and formed MES Business Unit. - Established a subsidiary "Testar Electronics", specialized in optoelectronic testing (LED, photodiode, VCSEL).
2009	- Acquired Raster Opto-Mechatronics, an optical measurement equipment maker.
2011	- Acquired Wise Life Technology, specialized in thermoelectric measurement systems.
2012	- Acquired QuadTech, a solution provider of electrical safety testers, passive component measurement testers, as well as AC and DC Power Sources, DC Loads and automation software.
2013	- Invested in ADIVIC Technology, a maker of RF measurement instruments.
2016	- Invested in Quantel Pte Ltd. as Chroma's Southeast Asia branch.
2017	- Established Innovative Nanotech, which builds nanoparticle monitoring systems for semiconductor manufacturing process control. - Invested in Touch Cloud, an AIoT solution provider.
2019	- Invested in Camtek, a global leading semiconductor metrology and inspection equipment vendor. - Formed Optical Inspection Solutions BU.
2022	- Acquired Environmental Stress Systems, whose main products consist of thermal forcing systems and could enable Chroma to grow the expertise in thermal control in semiconductor testing equipment. - Terminated Chroma New Material.

Source: Company data, Nomura research

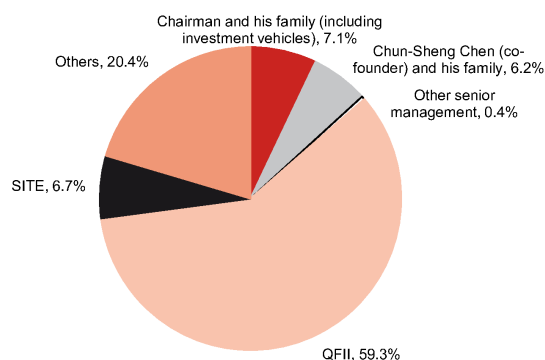
Fig. 59: Chroma's top shareholders

Name	Ownership	Remarks
Chun-Sheng Chen	3.6%	Co-founder
Grandcrest Holdings Inc.	3.5%	Shu-Chuan Chen (spouse of Leo Huang) is the representative; she also owns 1.01% of interest in Chroma.
Government of Singapore	3.5%	
Yu-Mei Hsueh	2.6%	Spouse of Chun-Sheng Chen
Leo Huang	2.6%	Chairman and co-founder
Labor Pension Fund	2.3%	
Yuanta/P-shares Taiwan Top 50 ETF	1.7%	
Norges Bank	1.4%	
Vanguard Total International Stock Index Fund	1.2%	
Vanguard Emerging Markets Stock Index Fund	1.2%	

Note: Data as of May 2026.

Source: Company data, Nomura research

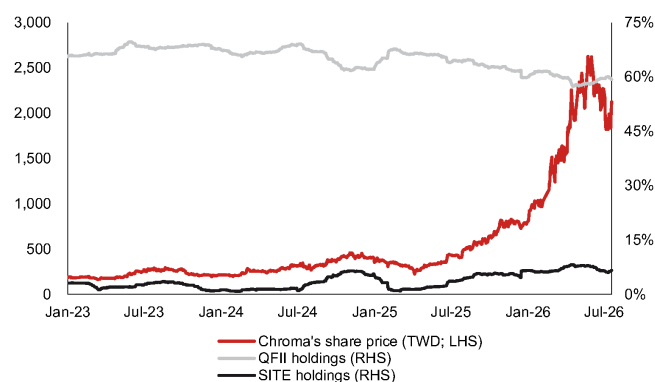
Fig. 60: Chroma's shareholder structure



Note: Data as of 22 July 2026.

Source: Company data, TEJ, Nomura research

Fig. 61: Chroma's share price vs. QFII and SITE ownerships



Note: Data as of 22 July 2026.

Source: TEJ, Nomura research

Fig. 62: Board of directors

Position	Name	Direct ownership	Background
Chairperson	Leo Huang	2.6%	- Electronics Engineering Department, National Chiao Tung University - Co-founder and CEO of Chroma
Director	I-Shih Tseng	0.1%	- Ph.D in Mechanical Engineering, Pennsylvania State University - Manager, Institute for Information Industry - Deputy Chief Engineer of Chroma - General Manager of Chroma
Director	David Yang	0.0%	- Electronics Engineering Department, National Chiao Tung University - Deputy General Manager of Chroma - General Manager of Chroma
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Independent director	Jia-Ruey Duann	0.0%	- Ph.D in Physics from North Dakota State University - Harvard Business School Advanced Management Program - Assistant Deputy General Manager, ITRI - CEO, ITRI Southern Region Campus/ITRI Central Region Campus - Director of Measurement Technology Development Center, ITRI - Adjunct Professo, Department of Physics, Chung Yuan Christian University - President of Automatic Optical Inspection Equipment Association - Assistant Researcher of Precision Instrument Development Center, National Science Council /Plant manager of optical devices
Independent director	Steven Wu	0.0%	- MBA, George Washington University Department of Industrial and Information Management, National Cheng Kung University - Senior Deputy General Manager, China Venture Management, Inc. - Senior Vice President, WI Harper Group - Deputy Manager, Investment Business Dept., Central Investment Holding Co., Ltd. - Assistant Manager, Investment Review Group, KMT Investment Business Management Committee - Senior Advisor, Corporate Financial Consulting Dept., Deloitte Taiwan - Investment Manager, DFV Fine Wines Group
Independent director	Janice Chang	0.0%	- Master of Business Administration, Major in Financial Management, Drexel University - Department of Accounting, National Chengchi University - VP, Capital Market Department, KGI Securities

Note: Data as of May 2026.

Source: Company data, Nomura research

Appendix A-1

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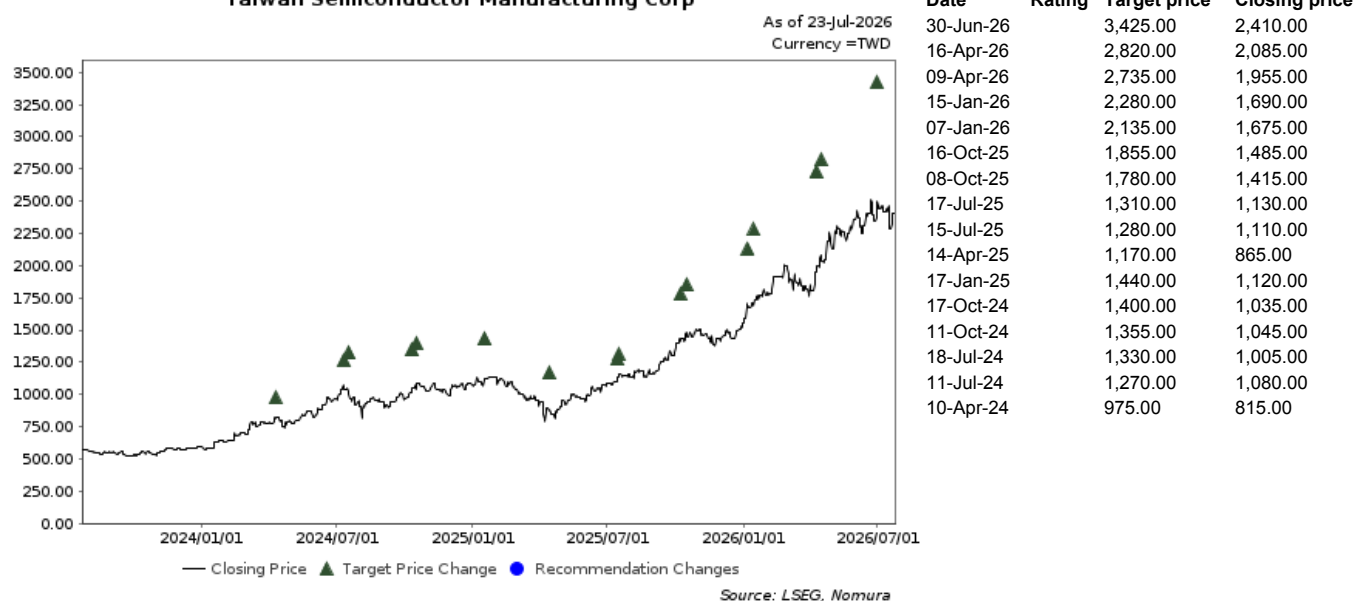
Issuer	Ticker	Price	Price date	Stock rating	Sector rating	Disclosures
Taiwan Semiconductor Manufacturing Corp	2330 TT	TWD 2,405.00	23-Jul-2026	Buy	N/A	
Chroma ATE Inc.	2360 TT	TWD 2,125.00	22-Jul-2026	Buy	N/A	

Taiwan Semiconductor Manufacturing Corp (2330 TT)

TWD 2,405.00 (23-Jul-2026) Buy (Sector rating: N/A)

Rating and target price chart (three year history)

Taiwan Semiconductor Manufacturing Corp



For explanation of ratings refer to the stock rating keys located after chart(s)

Valuation Methodology Our TP of TWD3,425 is based on 25x 2027F EPS. Our target P/E is at the higher end of historical range. The benchmark index for the stock is Taiwan TAIEX and SOX.

Risks that may impede the achievement of the target price Major downside risks are: 1) top-down macro issues because of US-China trade tensions; 2) weaker-than-expected sell-through compared with strong demand in the supply chain; 3) slower-than-expected technology migration; and 4) stronger-than-expected competition in advanced 5/3nm nodes.

Chroma ATE Inc. (2360 TT)

TWD 2,125.00 (22-Jul-2026) Buy (Sector rating: N/A)

Rating and target price chart (three year history)



For explanation of ratings refer to the stock rating keys located after chart(s)

Valuation Methodology Our TP of TWD2,845 is based on 38x average 2027-28F EPS, at the higher end of Chroma's historical trading band (14-58x). The benchmark of this stock is TAIEX.

Risks that may impede the achievement of the target price Major downside risks to our call include: 1) CoWoS and backend testing capacity expansion slowdown; 2) slower-than-expected product refresh, platform performance upgrade, or ramp-up by the AI chip end customers; 3) fiercer-than-expected from other international testing equipment makers; and 4) weaker-than-expected end-market demand, particularly for AI servers.

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As at 30 June 2026.

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